PES UNIVERSITY

Department of Computer Science & Engineering

Digital Forensics and Incident Responses

UE23CS342BA6

Assignment 2 Submission

Name of the Student	Pranav Hemanth
SRN	PES1UG23CS433
Section	G
Department	CSE
Campus	RR

Topic: Unit 1 - Lab 2

Digital Forensics and Incident Responses

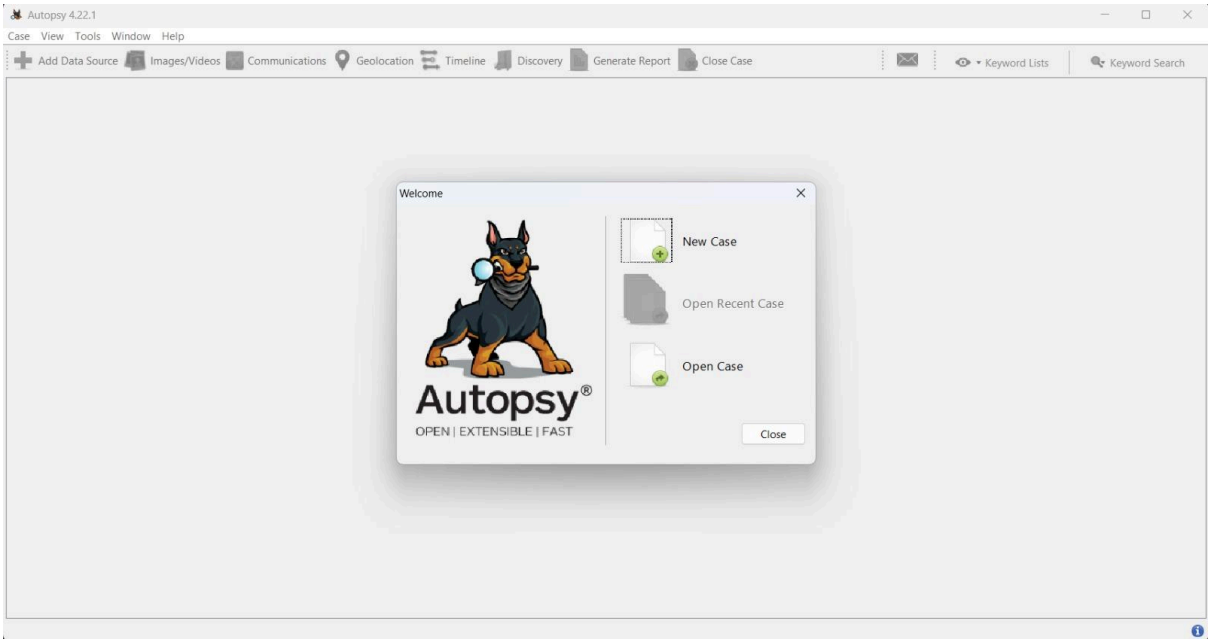
UE23CS342BA6

Analysis of Evidence Using Autopsy

Part A: Setup

1. Install Autopsy

- Download and install Autopsy from the official site.
- Launch Autopsy once installation completes.



2. Download the evidence image

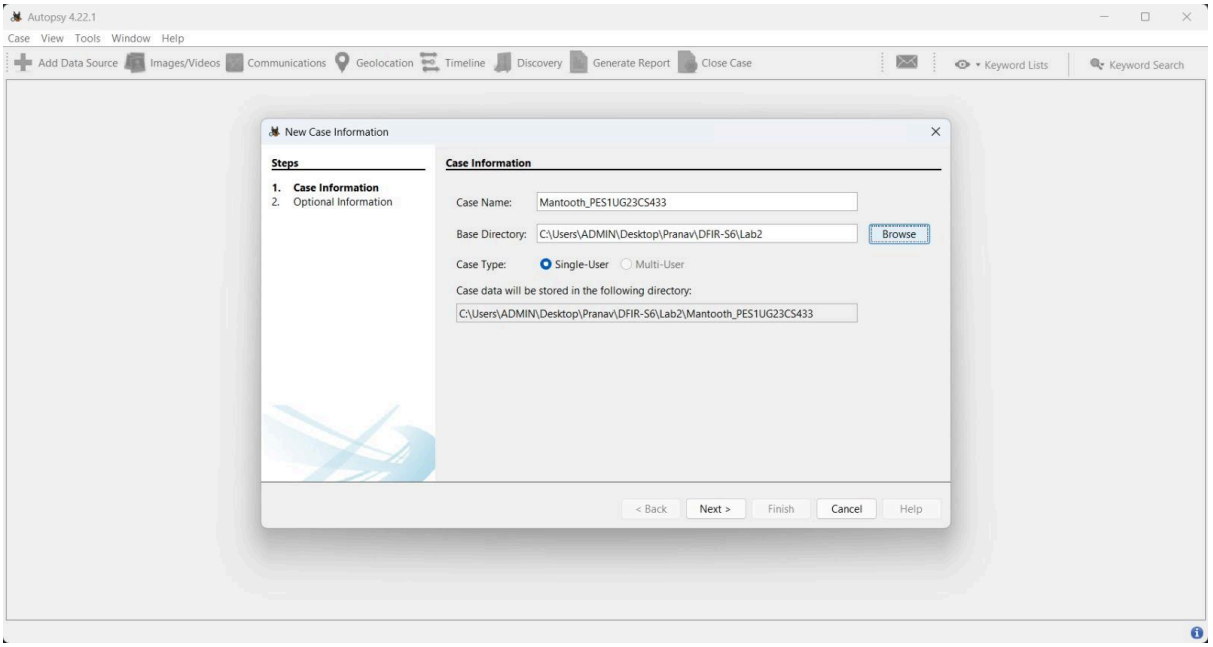
- Download Mantooth.E01 (this is your disk image for analysis).

```
pranavhemanth@Pranavs-MacBook-Pro-M3 DFIR-S6 %cd Lab2
pranavhemanth@Pranavs-MacBook-Pro-M3 Lab2 %ls -lh Mantooth.E01
-rw-rw-rw-@ 1 pranavhemanth  staff    38M Mar  5  2024 Mantooth.E01
pranavhemanth@Pranavs-MacBook-Pro-M3 Lab2 %
```

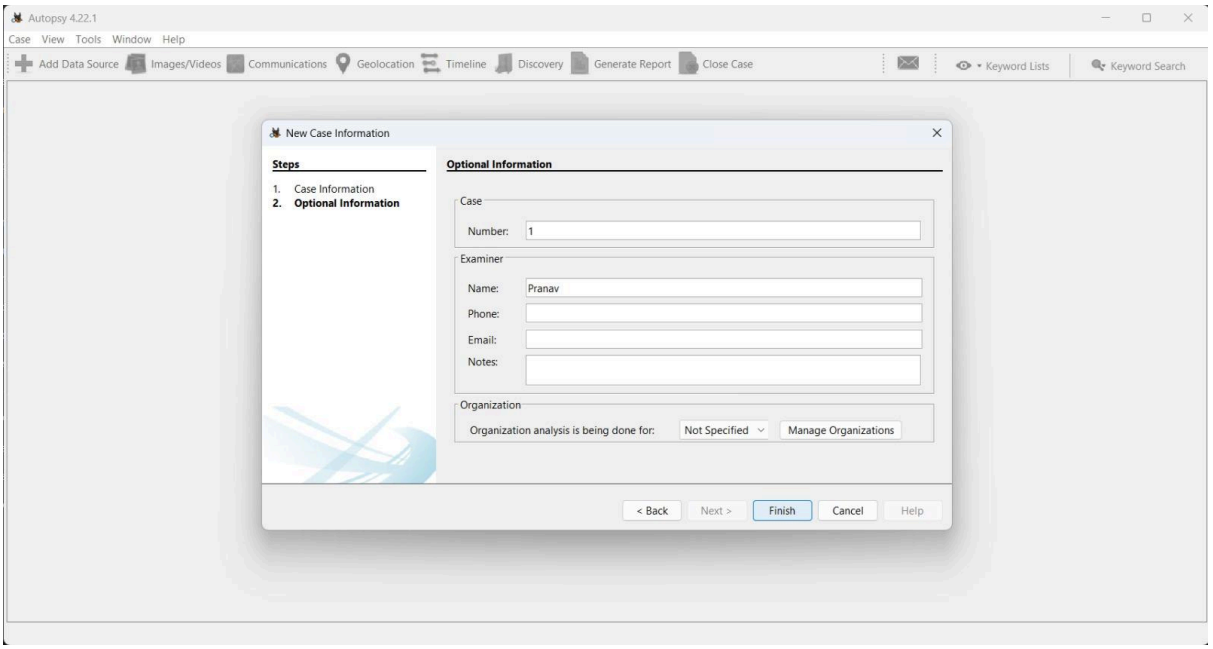
Part B: Create Case and Load Evidence

Step1: Create a new case

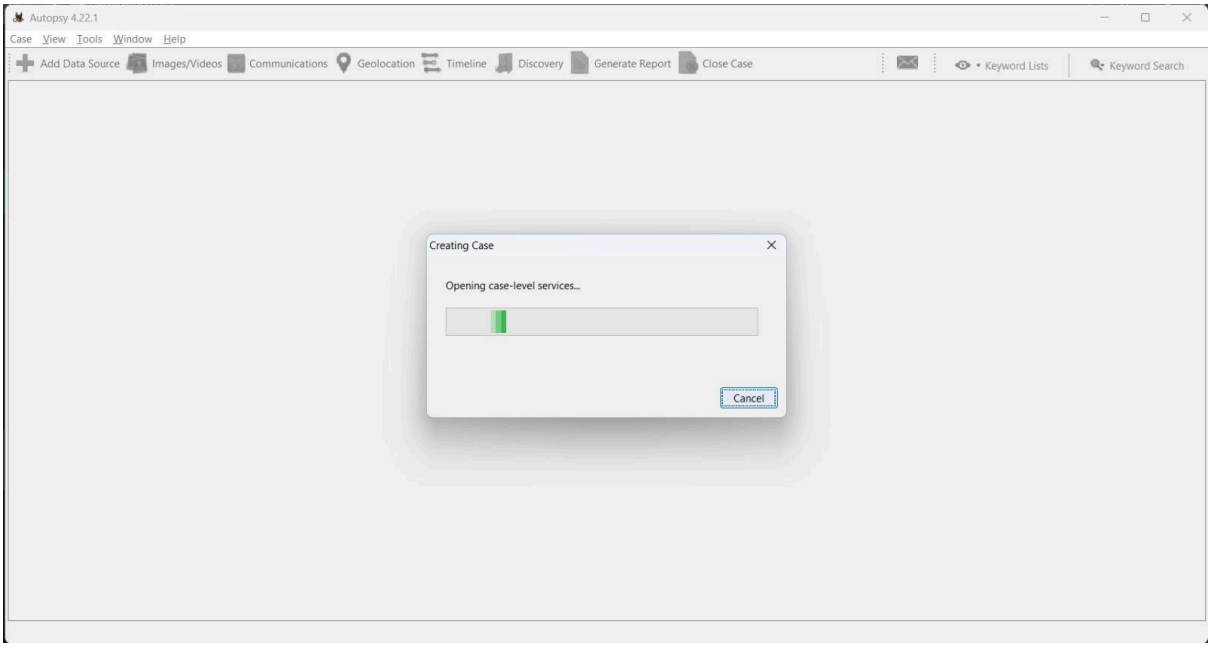
1) Open Autopsy → Create New Case
2) Case Name: anything (for example: Mantooth)
3) Choose a base directory (your lab folder)
4) Select Single-user



5) Enter Case Number (for example: 1) and your name as Examiner

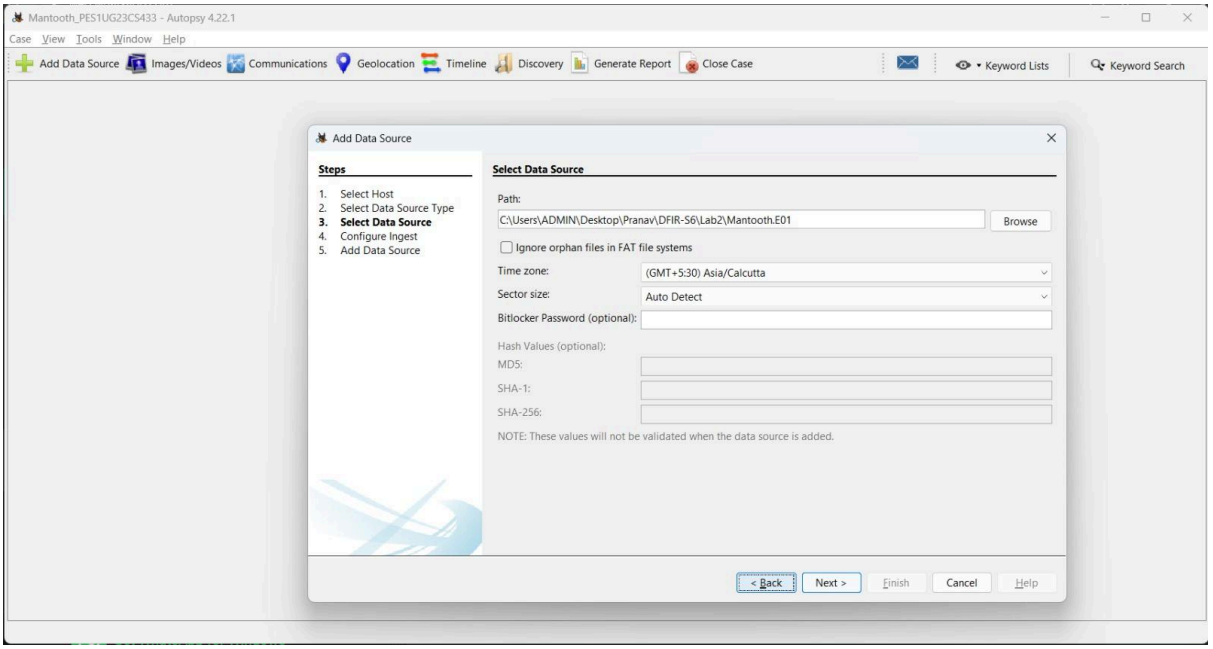


6) Finish

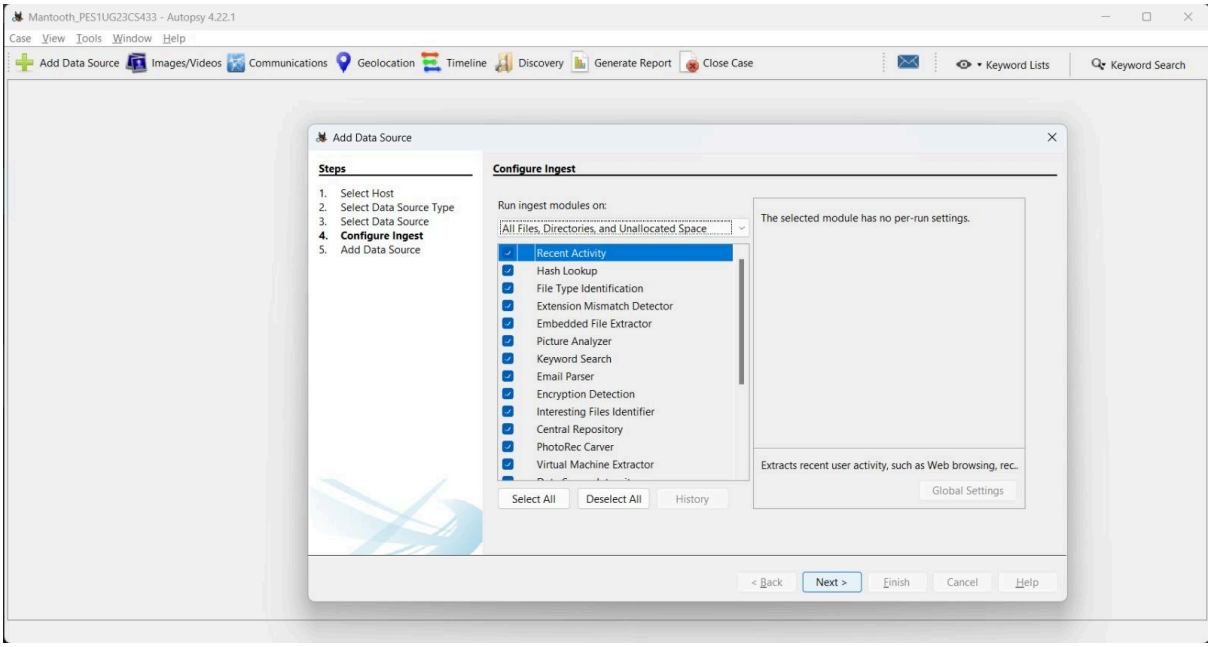


Step2: Add the data source

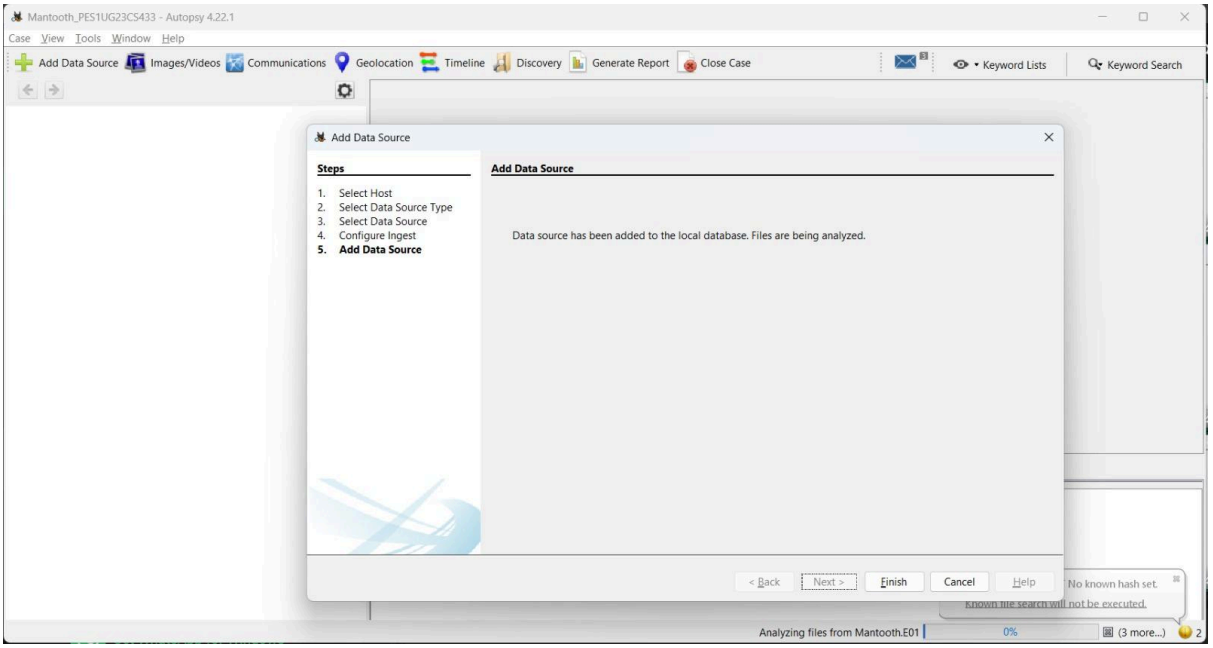
7) Select Disk Image or VM file
8) Browse and choose Mantooth.E01



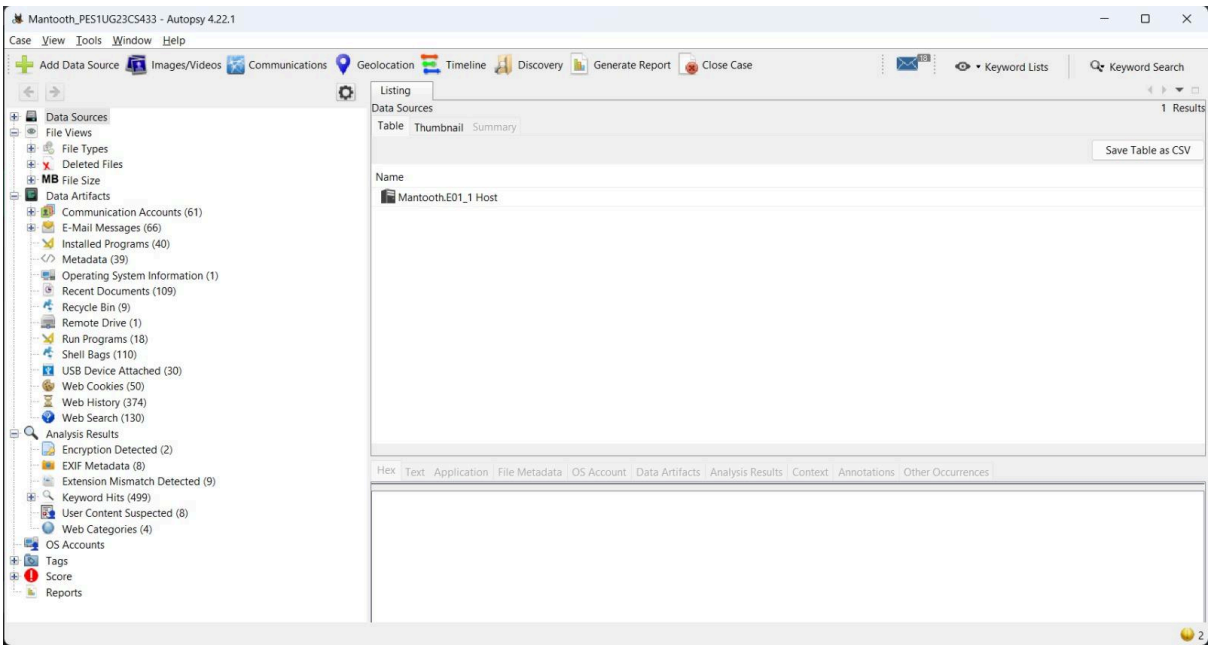
9) Keep default ingest modules enabled



10) Click Next → Finish

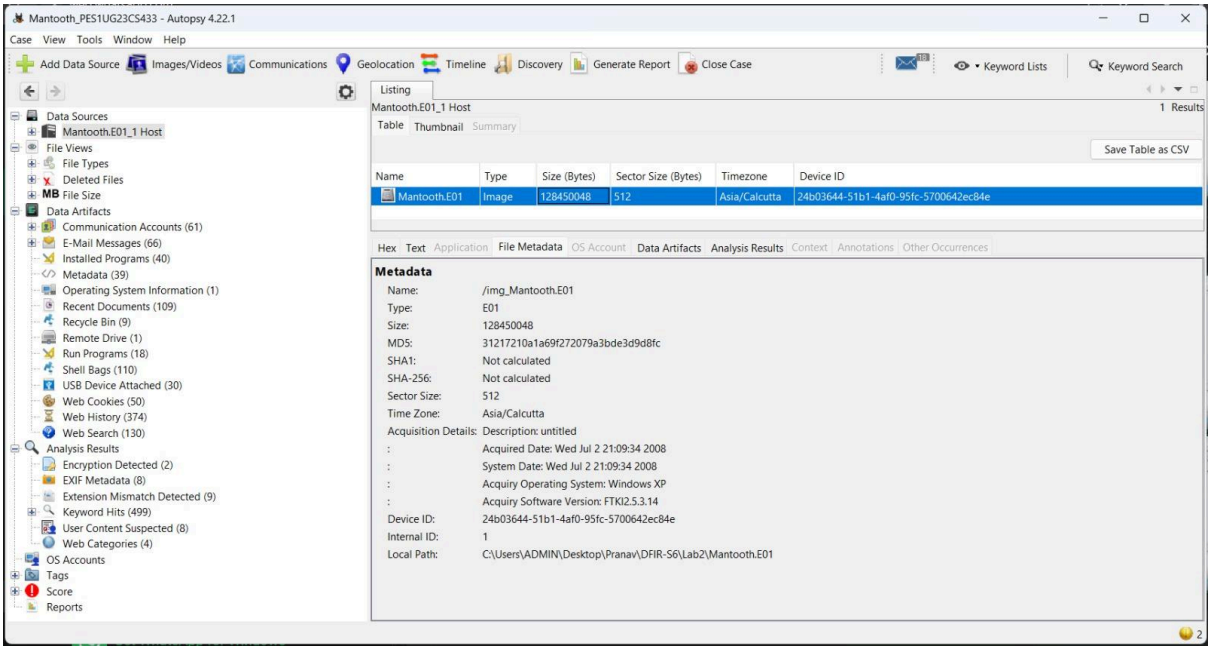


11) Wait for ingest to complete (important: let all modules run)

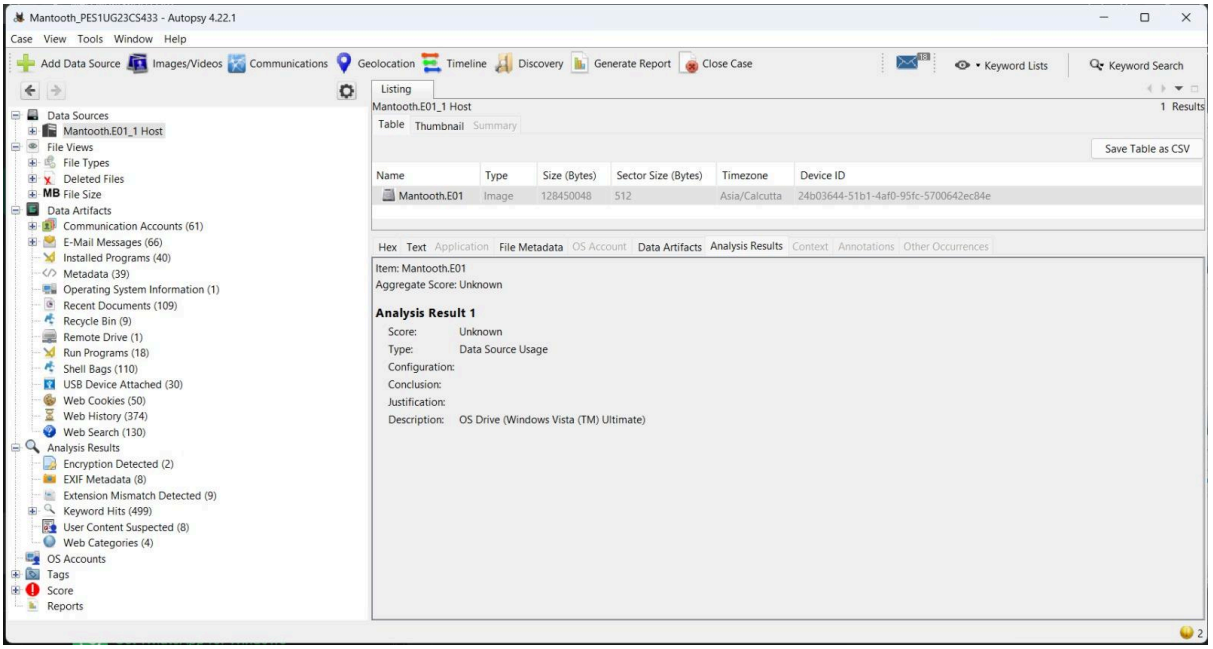


Part C: Answering the Lab Questions

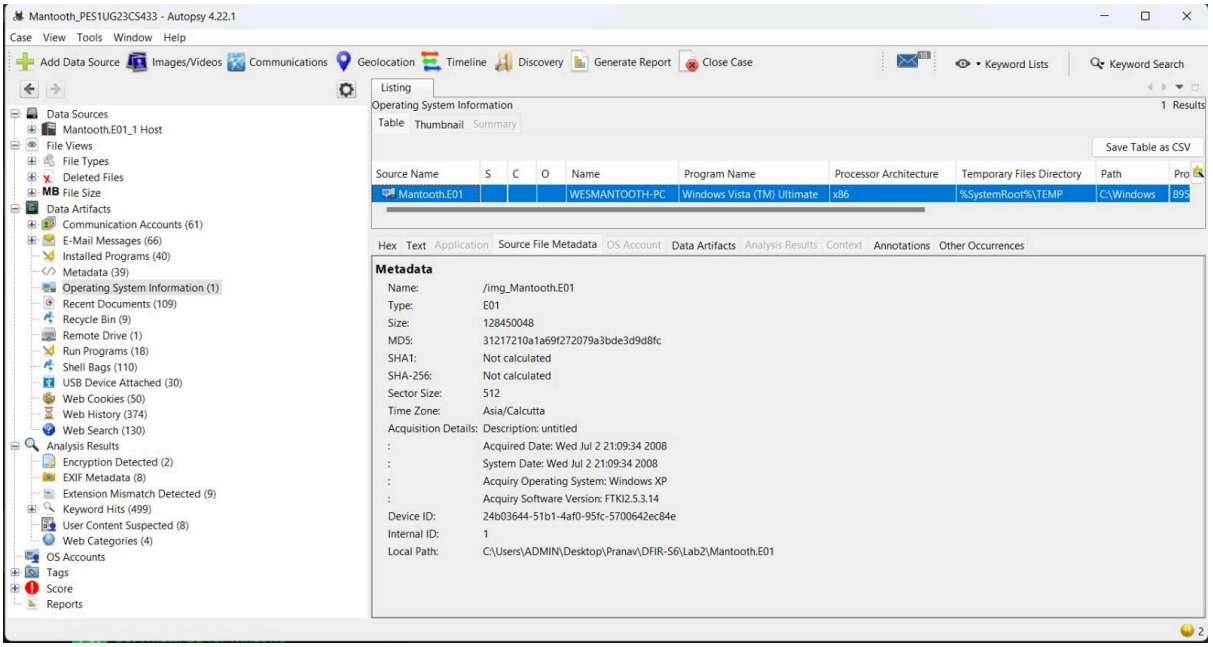
1. What is the image hash?
   Answer: The image (MD5) hash is 31217210a69f270279a3bde3d9d8fc



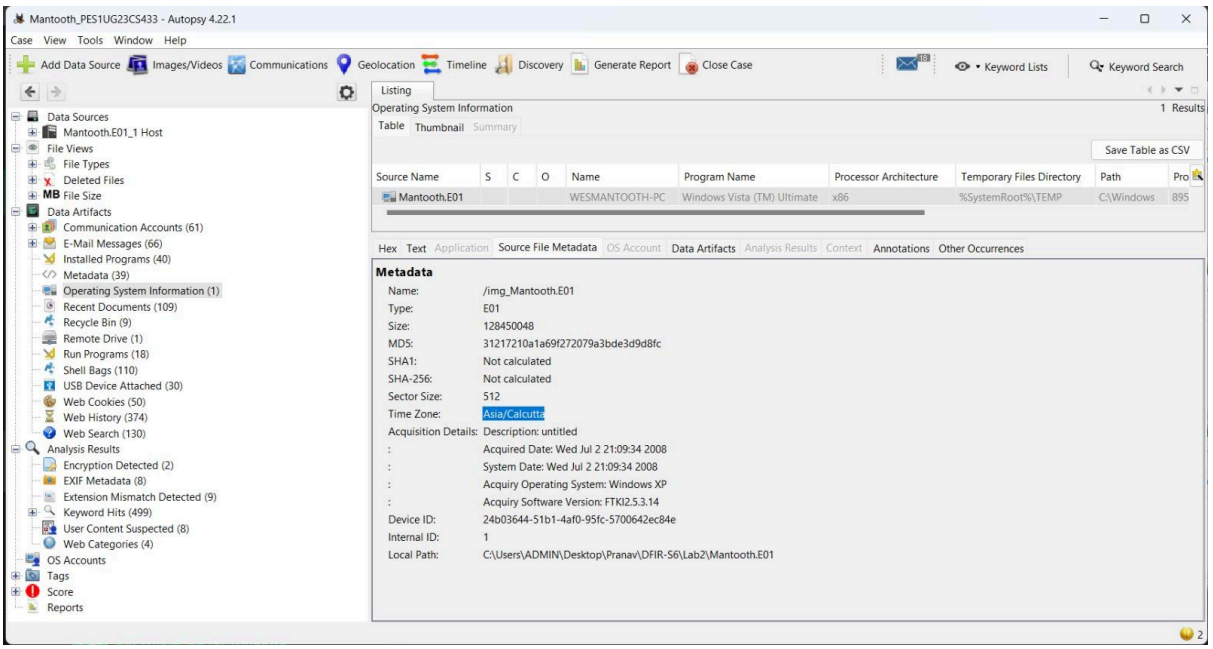
2. What operating system was used on the computer?
   Answer: The OS is Windows Vista Ultimate



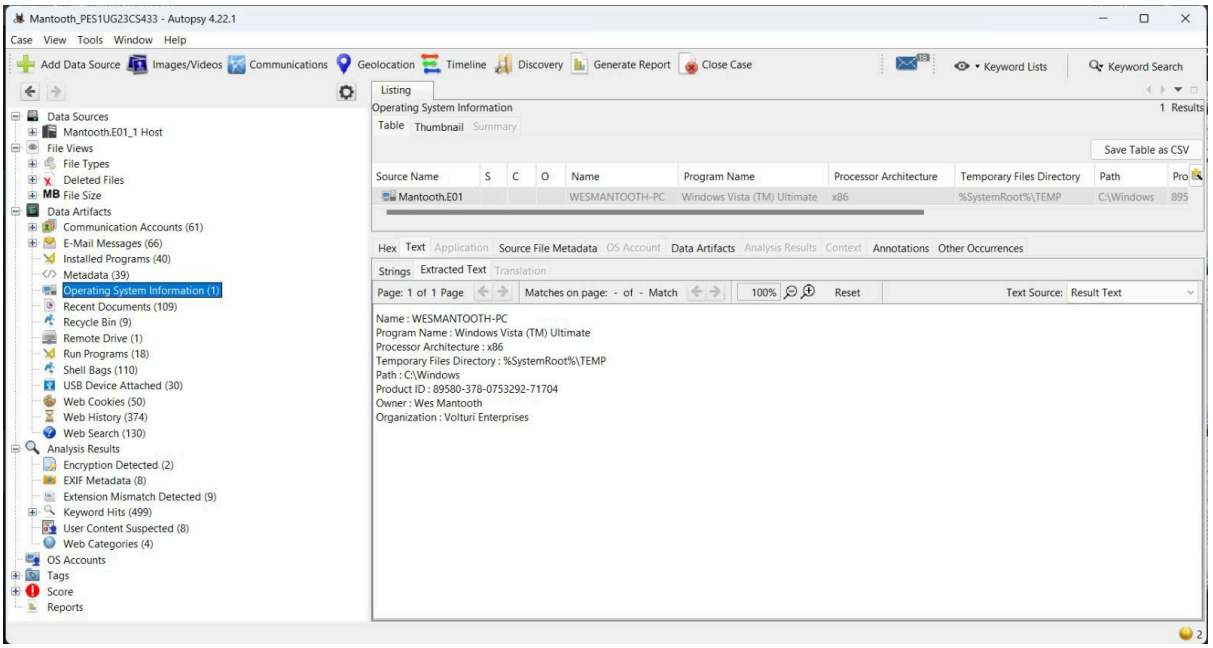
3. When was the install date?
   Answer: The Windows Vista Ultimate installation date is not directly listed in Autopsy's Operating System Information. However, analysis of the system activity timeline and the acquisition timestamp indicates that the operating system was installed around 2 July 2008.



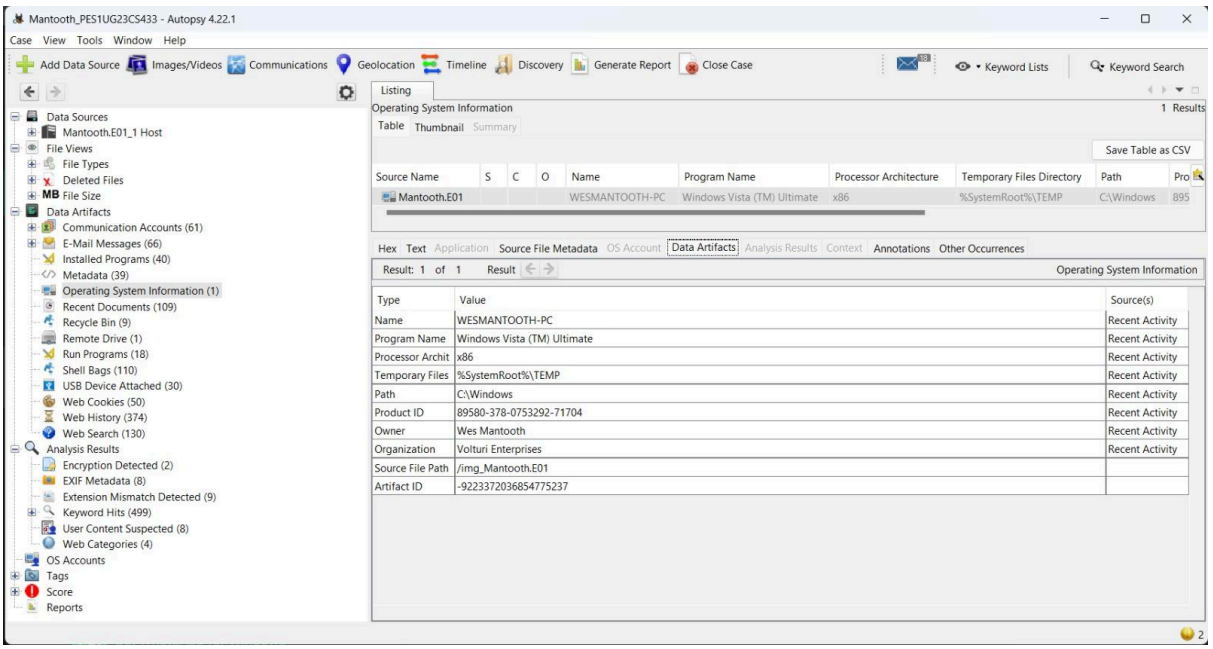
4. What are the timezone settings?
   Answer: The timezone is Asia/Calcutta



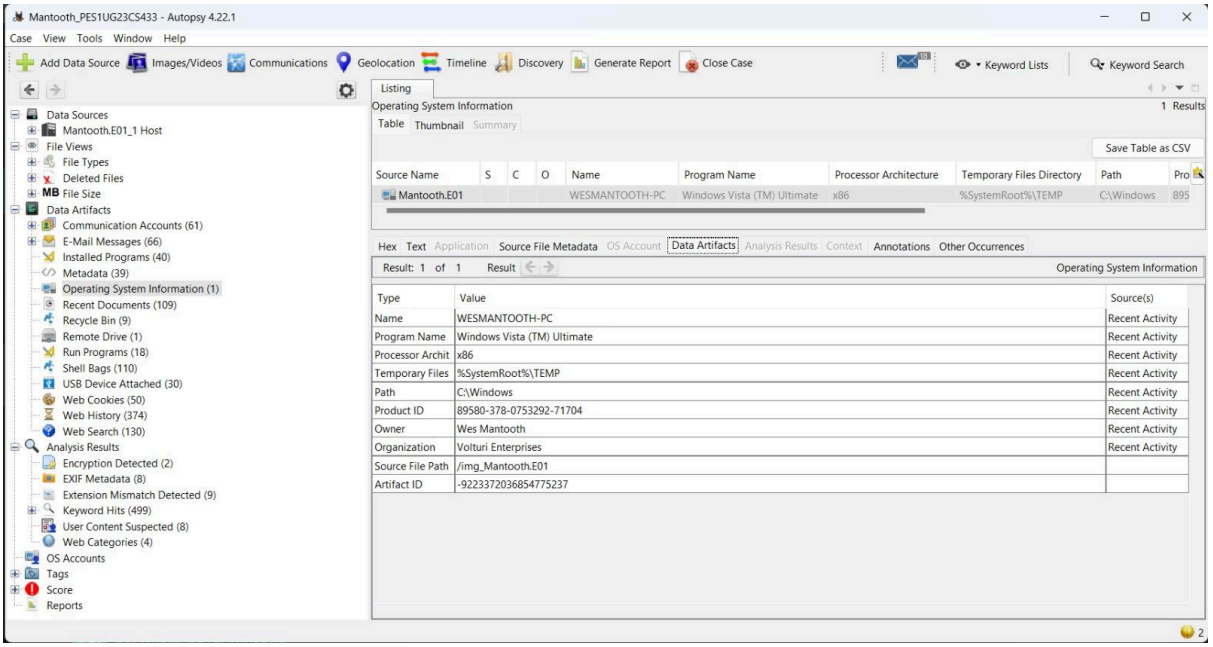
5. Who is the registered owner?
   Answer: The owner is Wes Mantooth



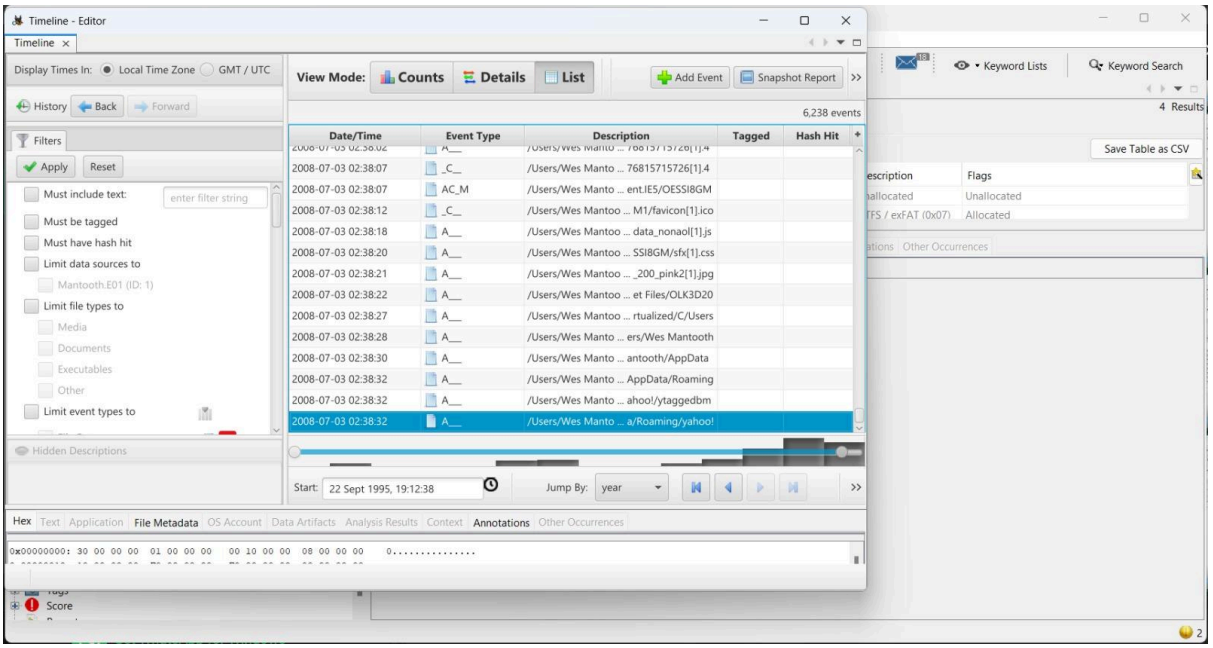
6. What is the computer account name?
   Answer: The computer account name is WESMANTOOTH-PC



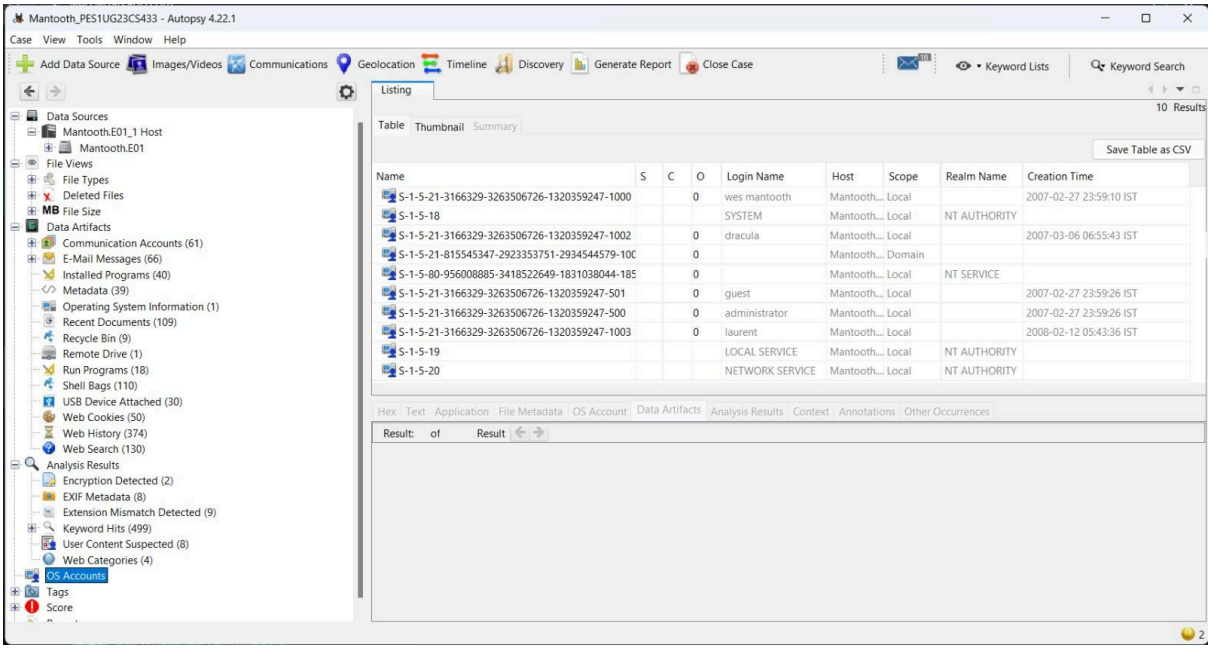
7. What is the primary domain name?
   Answer: The system is not joined to an Active Directory domain. As a standalone Windows system operating in a WORKGROUP environment, the effective primary domain resolves to the local machine name, which is WESMANTOOTH-PC.



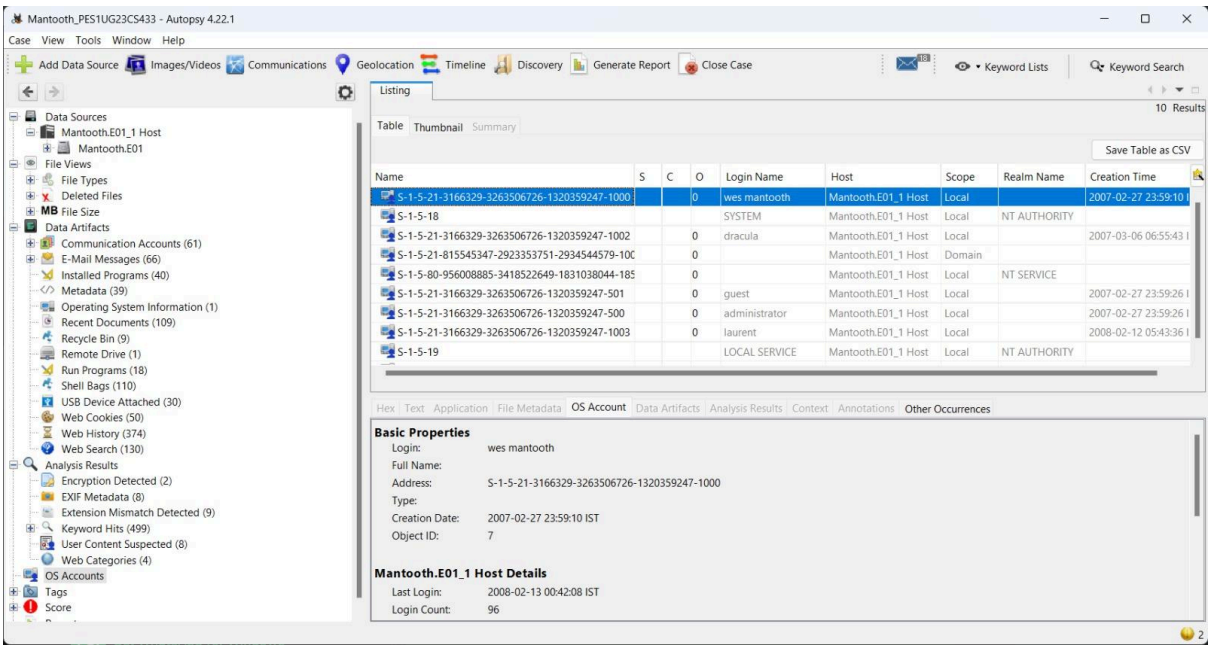
8. When was the last recorded computer shutdown date/time?
   Answer: Windows Event Log shutdown events were not available in the parsed artifacts. Therefore, the last recorded shutdown time was inferred from the system activity timeline. The final system-level activity occurred on **3rd July 2008 at 2:38:32 am**, indicating the last recorded shutdown of the computer.



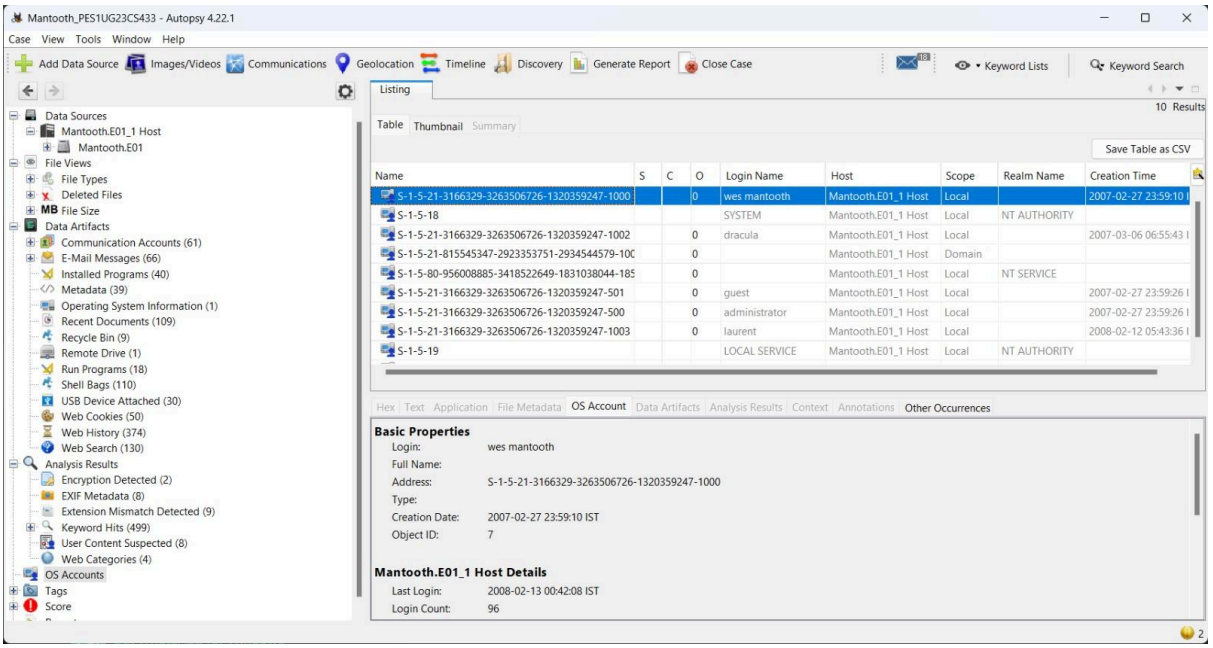
9. How many accounts are recorded (total number)?
   Answer:Autopsy identified a total of 10 accounts on the system. This count was obtained from the OS Accounts artifact, which lists all local and system-related user accounts present on the machine.



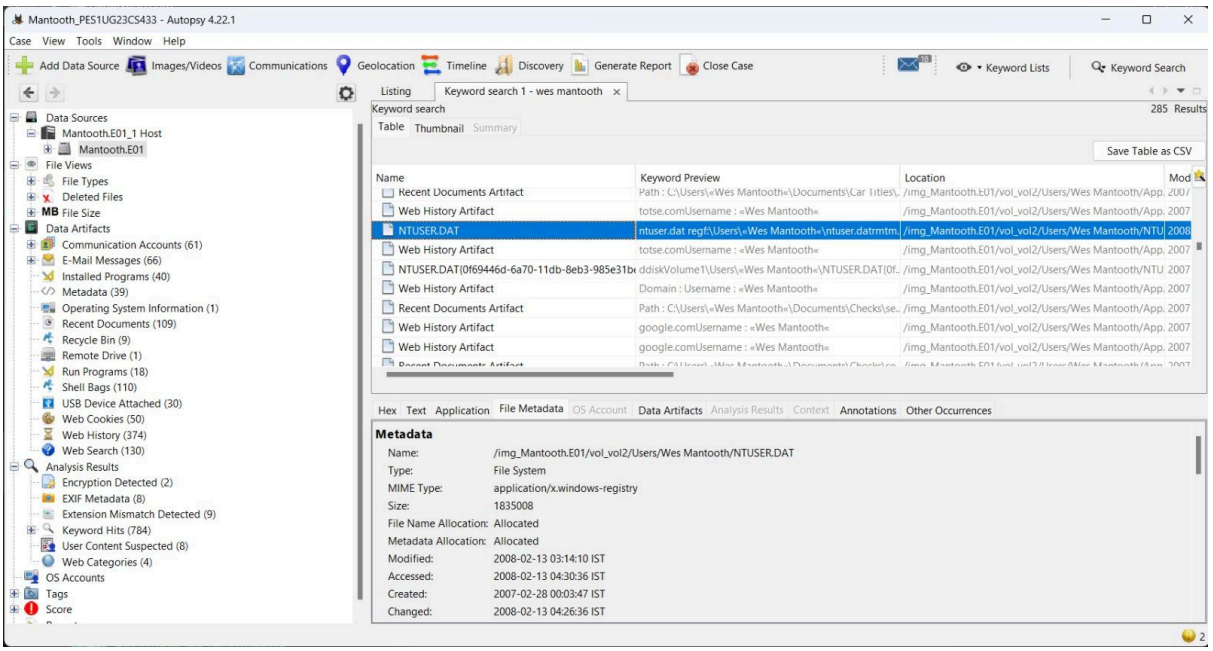
10. What is the account name of the user who mostly uses the computer?
    Answer: The account with the highest number of logins is wes mantooth, with a total of 96 login events.



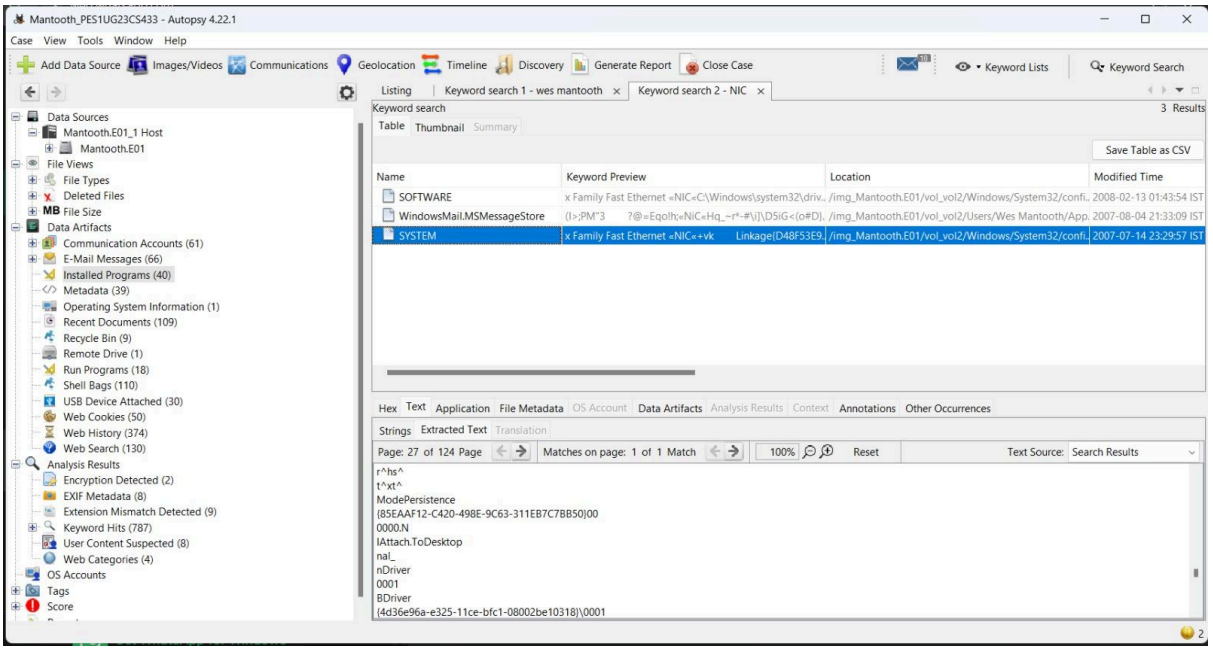
11. Who was the last user to logon to the computer?
    Answer: The last to login was wes mantooth at 00:42:08 on 13 of feb 2008



12. A search for the name of “Wes Mantooth” reveals multiple hits. One of these proves that Wes Mantooth is the administrator of this computer. What file is it?
    Answer: A keyword search for “Wes Mantooth” returned multiple results. One result corresponds to the user registry hive NTUSER.DAT, located in the user profile directory. This file is associated with the Wes Mantooth account and, when correlated with OS account information, confirms that Wes Mantooth is an administrator of the system.



13. List the network cards used by this computer
    Answer: The NIC is Realtek RTL8139/810x Family Fast Ethernet NIC

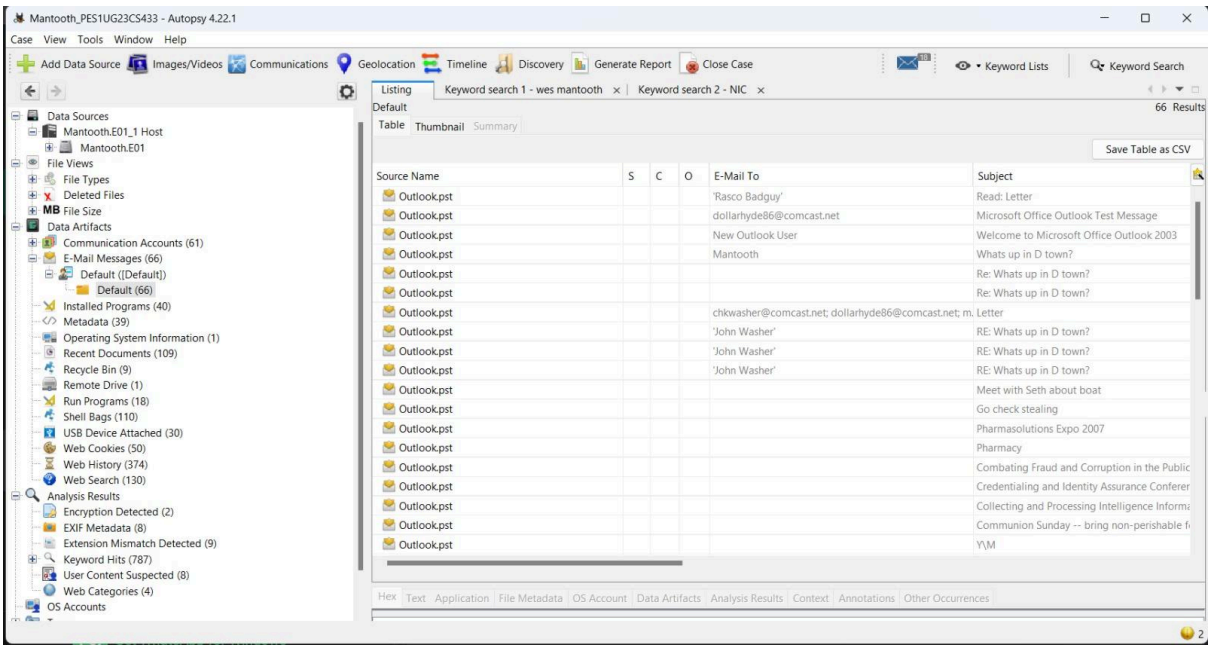


14. Find installed programs that may be used for Digital forensics/hacking.
    Answer: Analysis of the installed programs revealed the presence of software that may be used for digital forensics or security-related activities. Notably, AccessData FTK Imager v2.5.1 is a recognized digital forensics tool used for evidence acquisition. Additionally, tools such as TrueCrypt, VNC Free Edition, and FileZilla may be leveraged for data concealment, remote access, or file transfer, which are relevant in forensic and security investigations.

```
PS C:\Users\ADMIN\Desktop\Pranav\DFIR-S6\Lab2> cat '.\Installed Programs 20260130120602.csv'
"Source Name","Program Name",Date/Time,"Data Source"
SOFTWARE,WebEx,"2007-10-10 10:12:40 IST",Mantooth.E01
SOFTWARE,"FileZilla (remove only)","2007-06-24 00:23:53 IST",Mantooth.E01
SOFTWARE,"Microsoft Office Standard Edition 2003 v.11.0.5614.0","2007-04-17 23:25:28 IST",Mantooth.E01
SOFTWARE,"RTC Client API v1.2 v.1.2.0000","2007-04-17 21:43:27 IST",Mantooth.E01
SOFTWARE,"AccessData DNA 3 Worker v.3.3","2007-04-17 19:58:46 IST",Mantooth.E01
SOFTWARE,"AccessData Registry Viewer v.1.5","2007-04-14 00:01:22 IST",Mantooth.E01
SOFTWARE,QuickTime,"2007-04-13 23:36:51 IST",Mantooth.E01
SOFTWARE,"Adobe Reader 8 v.8.0.0","2007-04-12 23:26:59 IST",Mantooth.E01
SOFTWARE,"VNC Free Edition 4.1.2 v.4.1.2","2007-04-11 17:24:00 IST",Mantooth.E01
SOFTWARE,TrueCrypt,"2007-04-11 01:37:31 IST",Mantooth.E01
SOFTWARE,"Mozilla Firefox (2.0.0.3) v.2.0.0.3 (en-US)","2007-04-10 17:55:21 IST",Mantooth.E01
SOFTWARE,"AccessData FTK Imager v.2.5.1","2007-02-27 23:14:26 IST",Mantooth.E01
SOFTWARE,"BestCrypt 8.0","2007-02-27 23:08:21 IST",Mantooth.E01
SOFTWARE,"P2P Networking","2007-02-27 23:03:51 IST",Mantooth.E01
SOFTWARE,"AOL Uninstaller (Choose which Products to Remove)","2007-02-27 20:57:42 IST",Mantooth.E01
SOFTWARE,"Yahoo! Browser Services","2007-02-27 20:28:40 IST",Mantooth.E01
SOFTWARE,"Yahoo! Install Manager","2007-02-27 20:28:40 IST",Mantooth.E01
SOFTWARE,"Yahoo! Internet Mail","2007-02-27 20:28:36 IST",Mantooth.E01
SOFTWARE,"Yahoo! Toolbar","2007-02-27 20:28:31 IST",Mantooth.E01
SOFTWARE,"Yahoo! Toolbar","2007-02-27 20:28:29 IST",Mantooth.E01
SOFTWARE,"Yahoo! Messenger","2007-02-27 20:28:14 IST",Mantooth.E01
SOFTWARE,"WinRAR archiver","2007-02-27 20:01:25 IST",Mantooth.E01
SOFTWARE,Trillian,"2007-02-27 19:39:25 IST",Mantooth.E01
SOFTWARE,"Windows Live Messenger v.8.1.0178.00","2007-02-27 19:29:18 IST",Mantooth.E01
SOFTWARE,"Viewpoint Media Player","2007-02-27 19:21:07 IST",Mantooth.E01
SOFTWARE,"AIM 6","2007-02-27 19:21:04 IST",Mantooth.E01
SOFTWARE,AOLOCP_Y,"2007-02-27 19:20:53 IST",Mantooth.E01
SOFTWARE,DXM_Runtime,"2007-02-27 19:18:16 IST",Mantooth.E01
SOFTWARE,MPlayer2,"2007-02-27 19:18:16 IST",Mantooth.E01
SOFTWARE,"Adobe Flash Player 9 ActiveX v.9","2007-02-27 19:11:18 IST",Mantooth.E01
SOFTWARE,AddressBook,"2006-11-02 12:51:44 IST",Mantooth.E01
SOFTWARE,"Connection Manager","2006-11-02 12:51:44 IST",Mantooth.E01
SOFTWARE,DirectDrawEx,"2006-11-02 12:51:44 IST",Mantooth.E01
SOFTWARE,Fontcore,"2006-11-02 12:51:44 IST",Mantooth.E01
SOFTWARE,IE40,"2006-11-02 12:51:44 IST",Mantooth.E01
SOFTWARE,IE4Data,"2006-11-02 12:51:44 IST",Mantooth.E01
SOFTWARE,IE5BAKEX,"2006-11-02 12:51:44 IST",Mantooth.E01
SOFTWARE,IEData,"2006-11-02 12:51:44 IST",Mantooth.E01
SOFTWARE,MobileOptionPack,"2006-11-02 12:51:44 IST",Mantooth.E01
```

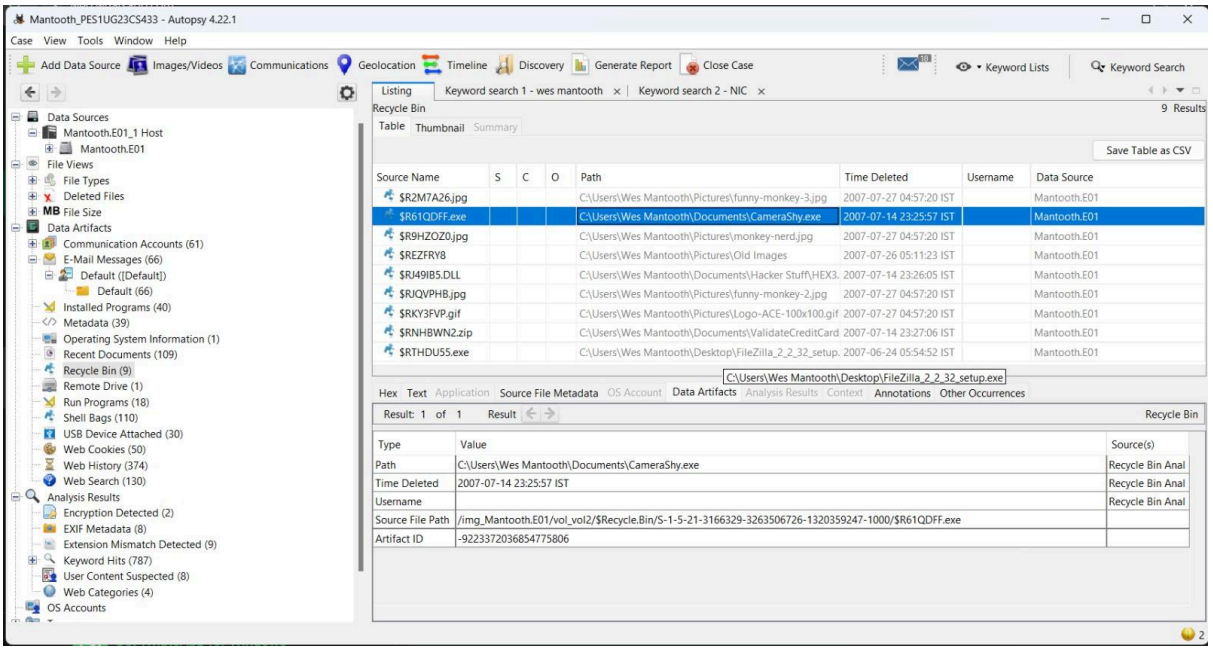
15. Which Email client is used by Mantooth?
    Answer: The primary email client used by Mantooth is Microsoft Office Outlook 2003, as confirmed by the presence of Outlook Personal Storage Table (.pst) artifacts. In addition, evidence of Yahoo Mail usage was also identified; however, this represents web-based email access rather than a dedicated email client. Therefore, Outlook 2003 is identified as the email client, while Yahoo Mail was accessed as a webmail service.

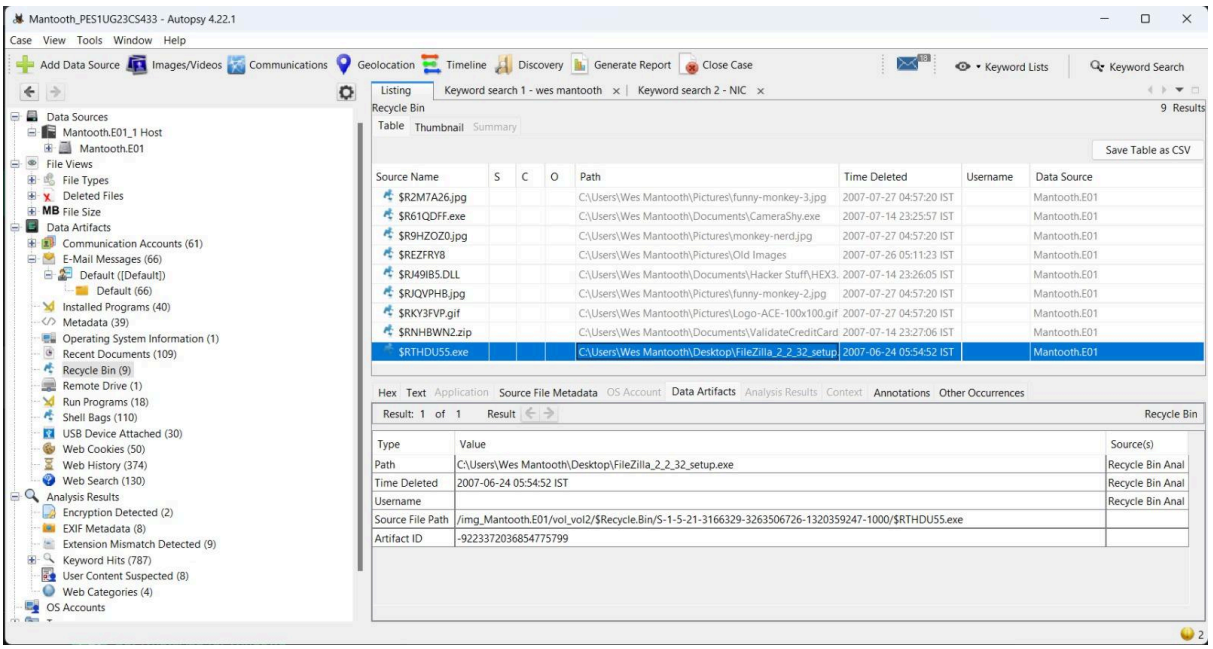
Jan - May 2025 Assignment SUBMISSION_UE23CS342BA6



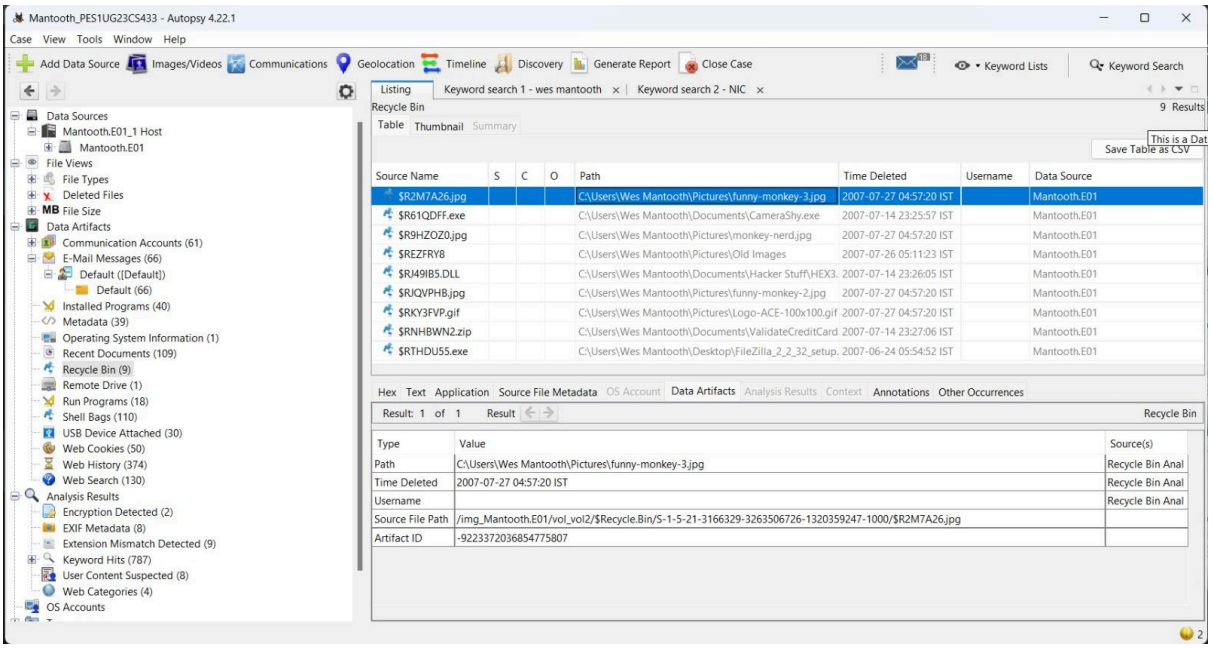
16. How many executable files are in the recycle bin?

    Answer: There are 2 executables in the recycle bin





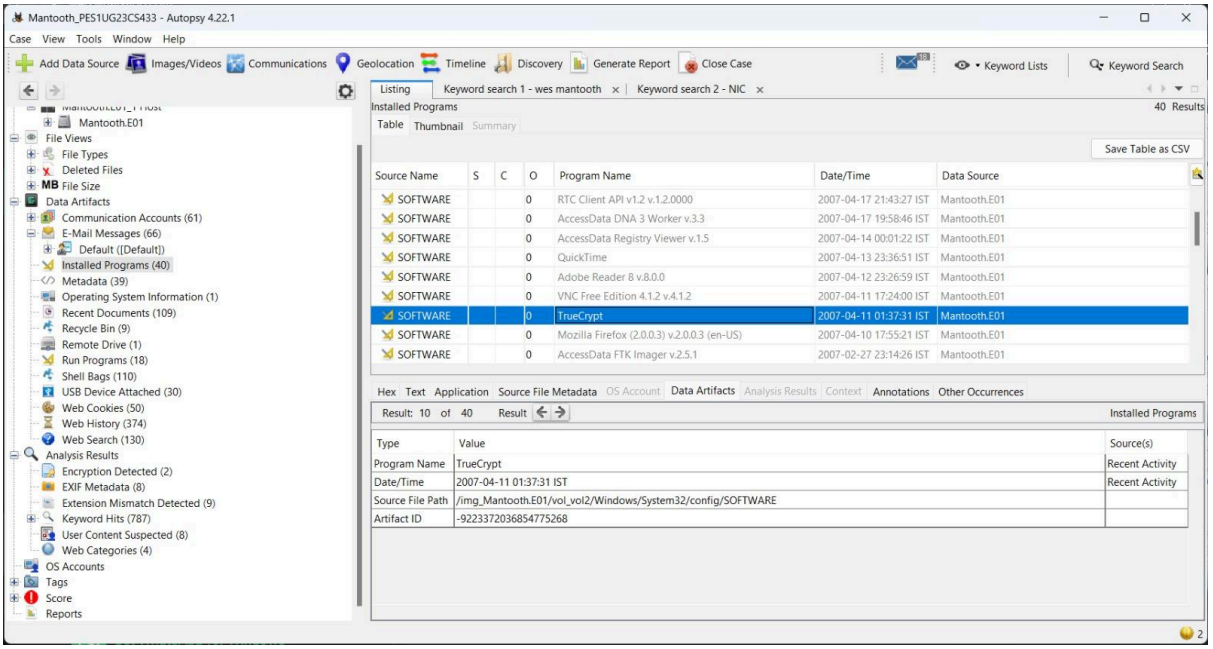
17. How many files are actually reported to be deleted by the file system?
    Answer: There are totally 9 files in the recycle bin



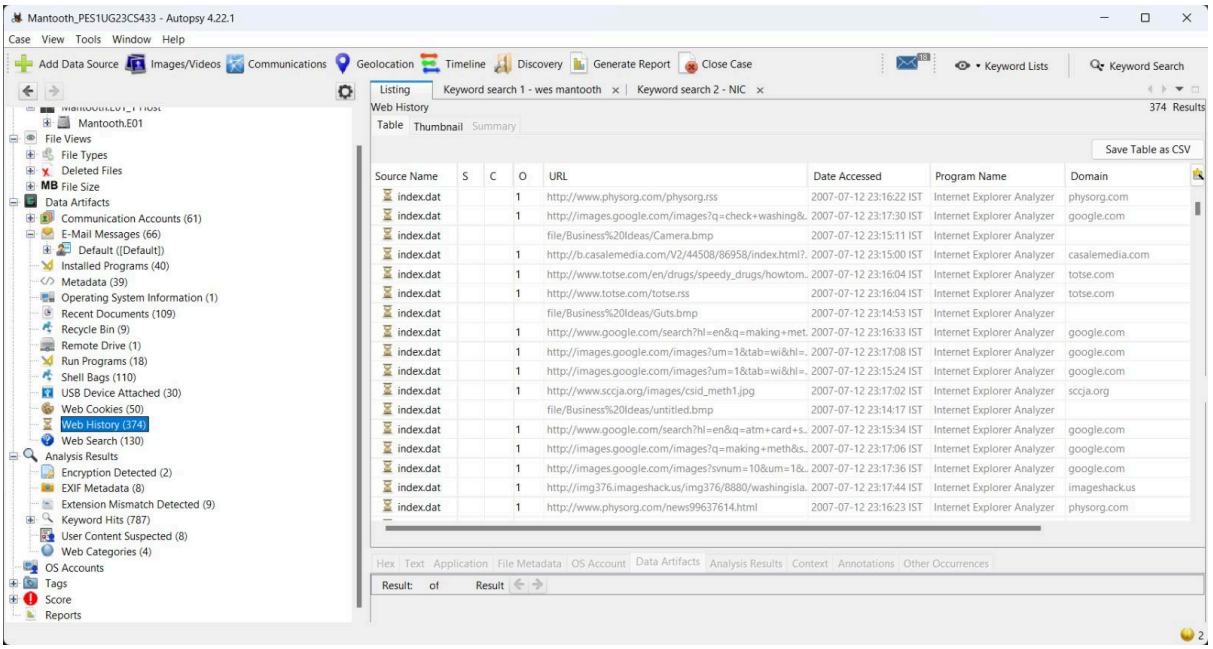
18. Are there any viruses on the computer?
    Answer: No evidence of viruses was identified on the computer. Autopsy did not report any hashset hits, malware indicators, or suspicious files during analysis. Based on the available forensic artifacts and automated analysis results, there is no indication that the system was infected with known viruses.

19. Is there encryption software installed on the Mantooth computer?
    Answer: Yes, encryption software is installed on the Mantooth computer. The Installed Programs artifacts show that TrueCrypt is present on the system.

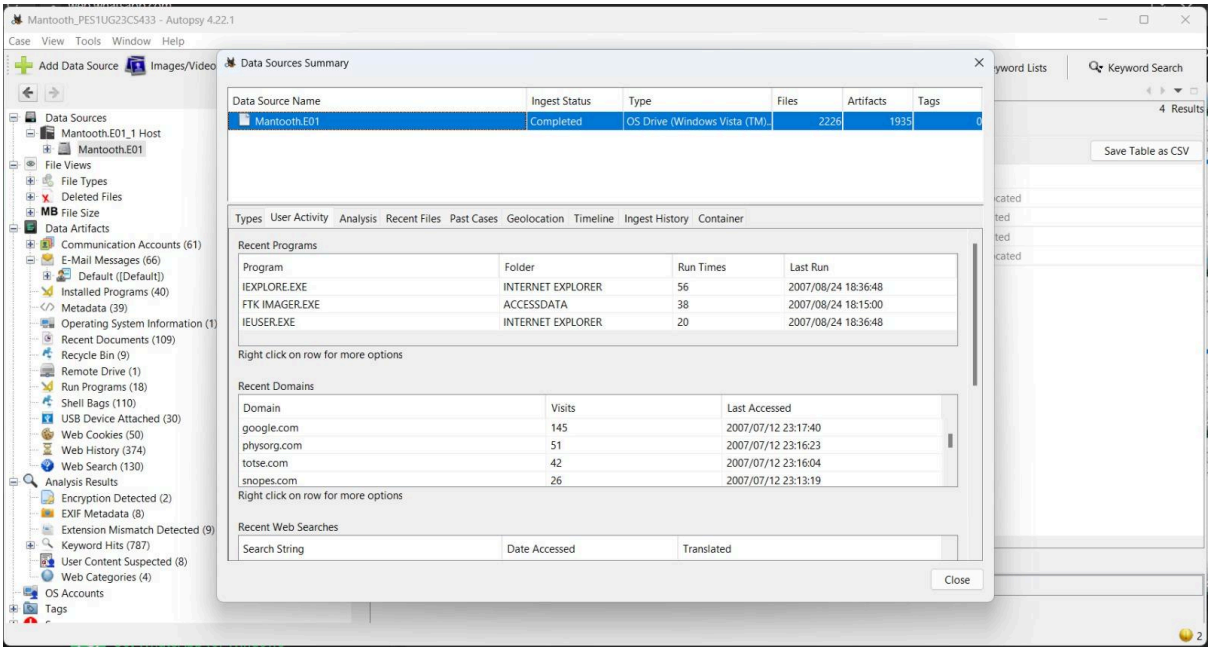
TrueCrypt is a disk and file encryption application used to protect data by encrypting storage volumes.



20. What is the most visited Internet domain and how many times it was visited ?
    Answer: The most visited internet domain is google.com, which was visited 145 times.



Jan - May 2025 Assignment SUBMISSION_UE23CS342BA6



RAM Dump Analysis using Volatility 3

Part A: Setup

1. Install Python (3.8 to 3.11, preferably 3.10)
2. Verify: python3 --version

```
pranavhemanth@Pranavs-MacBook-Pro-M3 Lab2 %python3 --version
Python 3.12.5
pranavhemanth@Pranavs-MacBook-Pro-M3 Lab2 %
```

3. Download Volatility 3

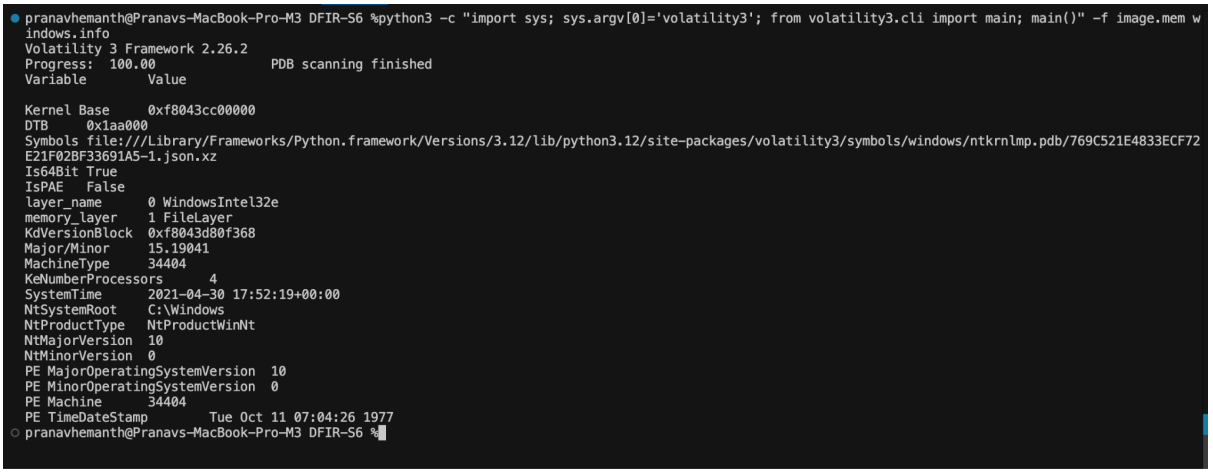
```
pranavhemanth@Pranavs-MacBook-Pro-M3 Lab2 %pip install volatility3
Requirement already satisfied: volatility3 in /Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages (2.26.2)
Requirement already satisfied: pefile>=2024.8.26 in /Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages (from volatility3) (2024.8.26)
pranavhemanth@Pranavs-MacBook-Pro-M3 Lab2 %
```

4. Verify Volatility 3

```
pranavhemanth@Pranavs-MacBook-Pro-M3 Lab2 %python3 -c "import sys; sys.argv[0]='volatility3'; from volatility3.cli import main; main()" -- -h
Volatility 3 Framework 2.26.2
usage: volatility3 [-h] [-c CONFIG] [--parallelism [{processes,threads,off}]] [-e EXTEND] [-p PLUGIN_DIRS] [-s SYMBOL_DIRS] [-v] [-l LOG]
                   [-o OUTPUT_DIR] [-q] [-r RENDERER] [-f FILE] [--write-config] [--save-config SAVE_CONFIG] [--clear-cache] [--cache-path CACHE_PATH]
                   [--offline | -u URL] [--filters FILTERS] [--hide-columns [HIDE_COLUMNS ...]] [--single-location SINGLE_LOCATION]
                   [--stackers [STACKERS ...]] [--single-swap-locations [SINGLE_SWAP_LOCATIONS ...]]
                   PLUGIN ...
volatility3: error: argument PLUGIN: invalid choice -- (choose from banners.Banners, configwriter.ConfigWriter, frameworkinfo.FrameworkInfo, isfinfo.IsfInfo, layerwriter.LayerWriter, linux.bash.Bash, linux.boottime.Boottime, linux.capabilities.Capabilities, linux.check_afinfo.Check_afinfo, linux.check_creds.Check_creds, linux.check_idt.Check_idt, linux.check_modules.Check_modules, linux.check_syscall.Check_syscall, linux.ebpf.EBPF, linux.elfs.Elfs, linux.envars.Envars, linux.graphics.fbdev.Fbdev, linux.hidden_modules.Hidden_modules, linux.iomem.IOMem, linux.ip.Addr, linux.ip.Link, linux.kallsyms.Kallsyms, linux.keyboard_notifiers.Keyboard_notifiers, linux.kmsg.Kmsg, linux.kthreads.Kthreads, linux.library_list.LibraryList, linux.lsmod.Lsmod, linux.lsof.Lsof, linux.malfind.Malfind, linux.malware.check_afinfo.Check_afinfo, linux.malware.check_creds.Check_creds, linux.malware.check_idt.Check_idt, linux.malware.check_modules.Check_modules, linux.malware.check_syscall.Check_syscall, linux.malware.hidden_modules.Hidden_modules, linux.malware.keyboard_notifiers.Keyboard_notifiers, linux.malware.malfind.Malfind, linux.malware.modxview.Modxview, linux.malware.netfilter.Netfilter, linux.malware.tty_check.Tty_Check, linux.module_extract.ModuleExtract, linux.modxview.Modxview, linux.mountinfo.MountInfo, linux.netfilter.Netfilter, linux.pagecache.Files, linux.pagecache.InodePages, linux.pagecache.RecoverFs, linux.pidhashtable.PIDHashTable, linux.proc.Maps, linux.psaux.PsAux, linux.pscallstack.PsCallStack, linux.pslist.PsList, linux.psscan.PsScan, linux.pstree.PsTree, linux.ptrace.Ptrace, linux.sockstat.Sockstat, linux.tracing.ftrace.CheckFtrace, linux.tracing.perf_events.PerfEvents, linux.tracing.tracepoints.CheckTracepoints, linux.tty_check.tty_check, linux.vmaregexscan.VmaRegExScan, linux.vmcoreinfo.VMCoreInfo, mac.bash.Bash, mac.check_syscall.Check_syscall, mac.check_sysctl.Check_sysctl, mac.check_trap_table.Check_trap_table, mac.dmesg.Dmesg, mac.ifconfig.Ifconfig, mac.kauth_listeners.Kauth_listeners, mac.kauth_scopes.Kauth_scopes, mac.kevents.Kevents, mac.list_files.List_Files, mac.lsmod.Lsmod, mac.lsof.Lsof, mac.malfind.Malfind, mac.mount.Mount, mac.netstat.Netstat, mac.proc_maps.Maps, mac.psaux.Psaux, mac.pslist.PsList, mac.pstree.PsTree, mac.socket_filters.Socket_filters, mac.timers.Timers, mac.trustedbsd.Trustedbsd, mac.vfsevents.VFSevents, regexscan.RegExScan, timeliner.Timeliner, vmscan.Vmscan, windows.amcache.Amcache, windows.bigpools.BigPools, windows.cachedump.Cachedump, windows.callbacks.Callbacks, windows.cmdline.CmdLine, windows.cmdscan.CmdScan, windows.consoles.Consoles, windows.crashinfo.Crashinfo, windows.debugregisters.DebugRegisters, windows.deskscan.DeskScan, windows.desktops.Desktops, windows.devicetree.DeviceTree, windows.dlllist.DllList, windows.driverirp.DriverIrp, windows.drivermodule.DriverModule, windows.driverscan.DriverScan, windows.dumpfiles.DumpFiles, windows.envars.Envars, windows.etwpatch.EtwPatch, windows.filescan.FileScan, windows.getservicesids.GetServiceSIDs, windows.getsids.GetSIDs, windows.handles.Handles, windows.hashdump.Hashdump, windows.hollowprocesses.HollowProcesses, windows.iat.IAT, windows.info.Info, windows.joblinks.JobLinks, windows.kpcrs.KPCRs, windows.ldrmodules.LdrModules, windows.lsadump.Lsadump, windows.malfind.Malfind, windows.malware.drivermodule.DriverModule, windows.malware.hollowprocesses.HollowProcesses, windows.malware.ldrmodules.LdrModules, windows.malware.malfind.Malfind, windows.malware.processghosting.ProcessGhosting, windows.malware.psxview.PsXView, windows.malware.skeleton_key_check.Skeleton_Key_Check, windows.malware.suspicious_threads.SuspiciousThreads, windows.malware.svcdiff.SvcDiff, windows.malware.unhooked_system_calls.UnhookedSystemCalls, windows.mbrscan.MBRScan, windows.memmap.Memmap, windows.modscan.ModScan, windows.modules.Modules, windows.mutantscan.MutantScan, windows.netscan.NetScan, windows.netstat.NetStat, windows.orphan_kernel_threads.Threads, windows.pe_symbols.PESymbols, windows.pedump.PEDump, windows.poolscanner.PoolScanner, windows.privileges.Privs, windows.processghosting.ProcessGhosting, windows.pslist.PsList, windows.psscan.PsScan, windows.pstree.PsTree, windows.psxview.PsXView, windows.registry.amcache.Amcache, windows.registry.cachedump.Cachedump, windows.registry.certificates.Certificates, windows.registry.getcellroutine.GetCellRoutine, windows.registry.hashdump.Hashdump, windows.registry.hivelist.HiveList, windows.registry.hivescan.HiveScan, windows.registry.lsadump.Lsadump, windows.registry.printkey.PrintKey, windows.registry.scheduled_tasks.ScheduledTasks, windows.registry.userassist.UserAssist, windows.scheduled_tasks.ScheduledTasks, windows.sessions.Sessions, windows.shimcachemem.ShimcacheMem, windows.skeleton_key_check.Skeleton_Key_Check, windows.ssdt.SSDT, windows.statistics.Statistics, windows.strings.Strings, windows.suspended_threads.SuspendedThreads, windows.suspicious_threads.SuspiciousThreads, windows.svcdiff.SvcDiff, windows.svclist.SvcList, windows.svcscan.SvcScan, windows.symlinkscan.SymlinkScan, windows.thrdscan.ThrdScan, windows.threads.Threads, windows.timers.Timers, windows.truecrypt.Passphrase, windows.unhooked_system_calls.unhooked_system_calls, windows.unloadedmodules.UnloadedModules, windows.vadinfo.VadInfo, windows.vadregexscan.VadRegExScan, windows.vadwalk.VadWalk, windows.verinfo.VerInfo, windows.virtmap.VirtMap, windows.windows.Windows, windows.windowstations.WindowStations)
pranavhemanth@Pranavs-MacBook-Pro-M3 Lab2 %
```

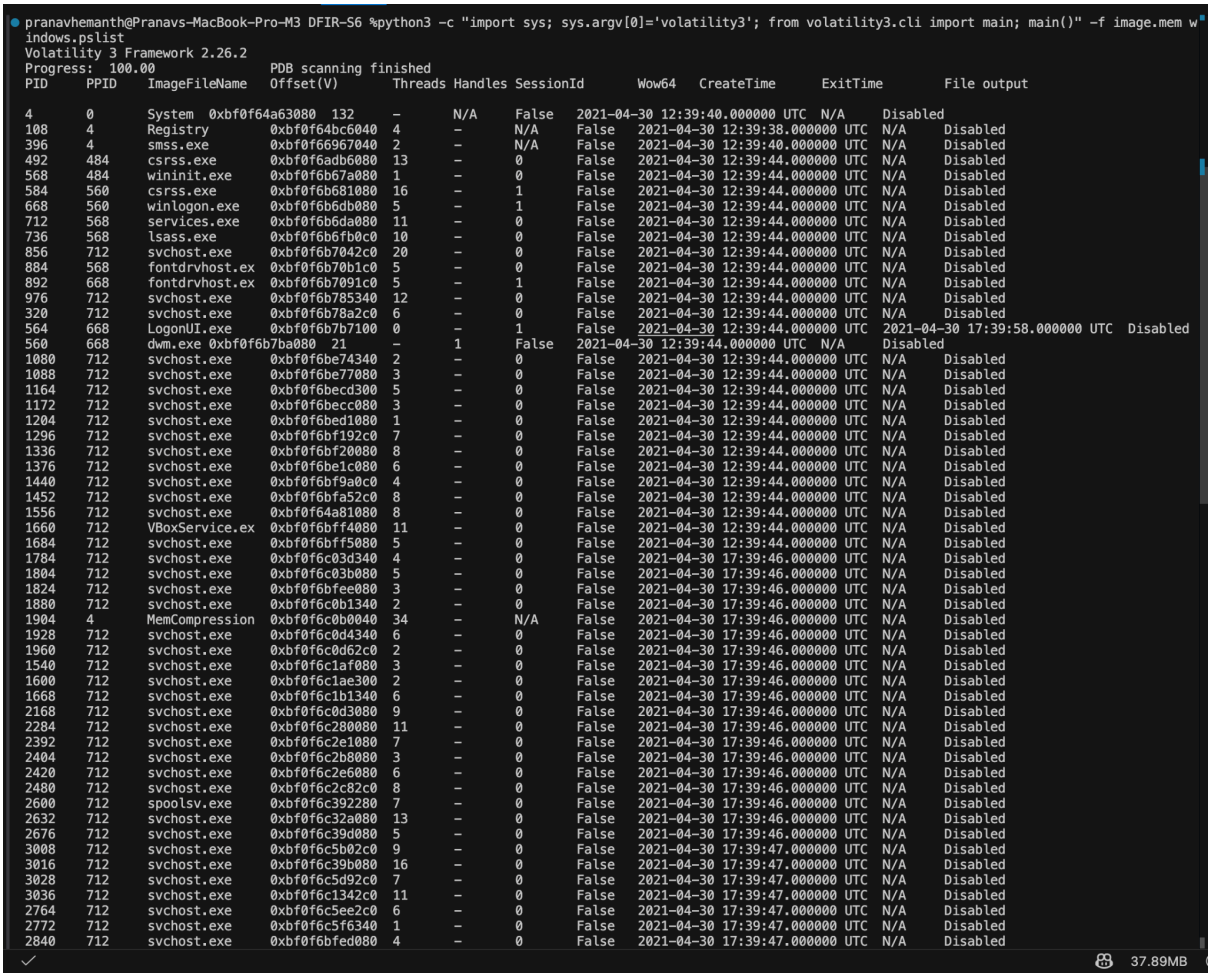
Part B: Core Analysis Workflow

1. Identify Operating System and Kernel Details
   Question: How can you determine the operating system version, kernel build, and system architecture of the given memory dump using Volatility 3?
   Answer:
   Command: python3 -c "import sys; sys.argv[0]='volatility3'; from volatility3.cli import main; main()" -f image.mem windows.info



The operating system version, kernel build, and system architecture of the memory dump were identified using Volatility 3's windows.info plugin. This plugin analyzes kernel structures within the memory image to extract system metadata such as the Windows version, kernel build number, and architecture. The information was obtained by executing the windows.info plugin against the memory dump using Volatility 3.

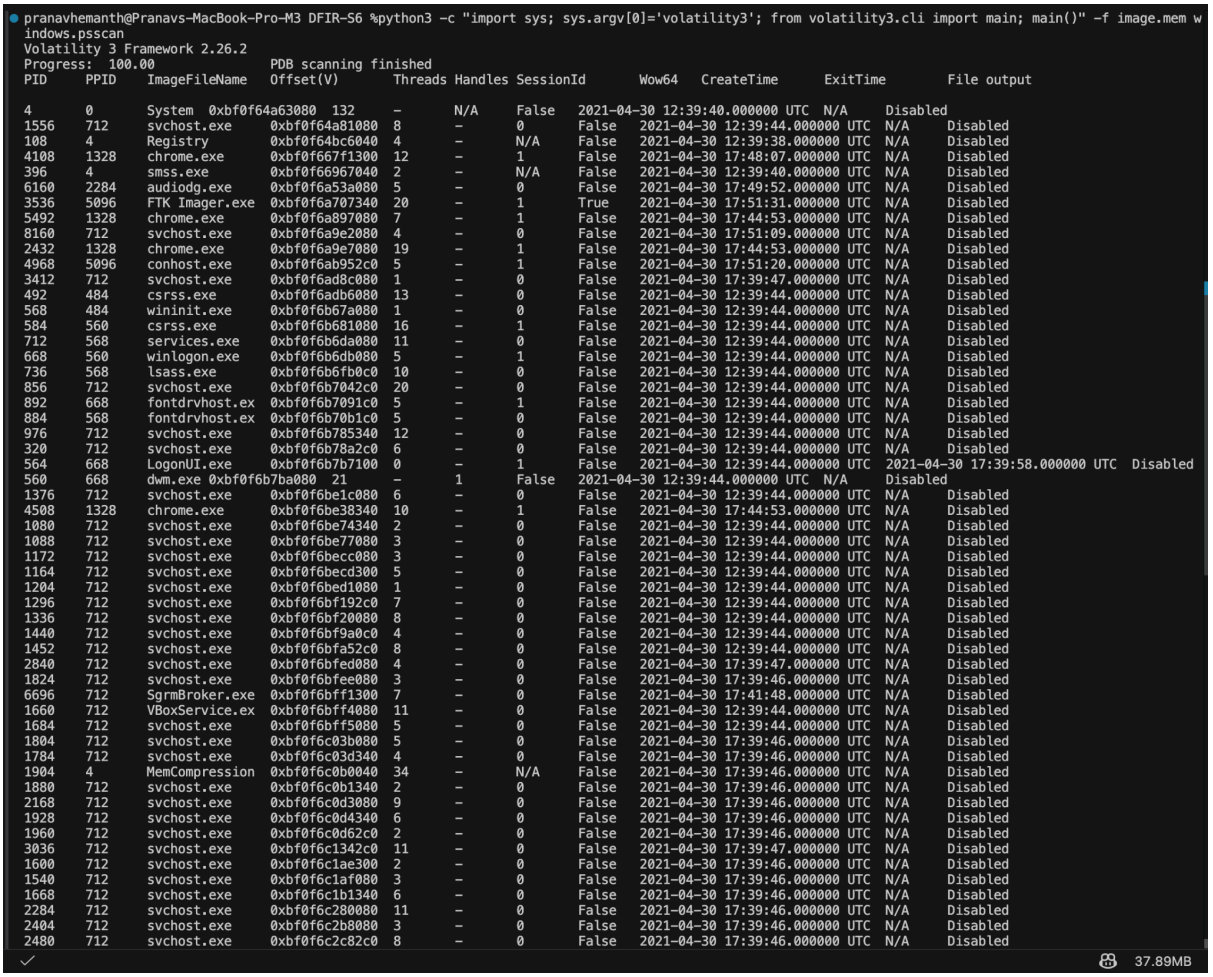
2. List Active Running Processes
   Question: How can you identify all active processes running on the system at the time the memory image was acquired?
   Answer:
   Command: python3 -c "import sys; sys.argv[0]='volatility3'; from volatility3.cli import main; main()" -f image.mem windows.pslist



```
pranavhemanth@Pranavs-MacBook-Pro-M3 DFIR-S6 %python3 -c "import sys; sys.argv[0]='volatility3'; from volatility3.cli import main; main()" -f image.mem w
indows.pslist
Volatility 3 Framework 2.26.2
Progress:  100.00               PDB scanning finished
PID     PPID    ImageFileName   Offset(V)       Threads Handles SessionId       Wow64   CreateTime      ExitTime        File output

4       0       System  0xbf0f64a63080  132     -       N/A     False   2021-04-30 12:39:40.000000 UTC  N/A     Disabled
108     4       Registry        0xbf0f64bc6040  4       -       N/A     False   2021-04-30 12:39:38.000000 UTC  N/A     Disabled
396     4       smss.exe        0xbf0f66967040  2       -       N/A     False   2021-04-30 12:39:40.000000 UTC  N/A     Disabled
492     484     csrss.exe       0xbf0f6adb6080  13      -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
568     484     wininit.exe     0xbf0f6b67a080  1       -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
584     560     csrss.exe       0xbf0f6b681080  16      -       1       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
668     560     winlogon.exe    0xbf0f6b6db080  5       -       1       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
712     568     services.exe    0xbf0f6b6da080  11      -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
736     568     lsass.exe       0xbf0f6b6fb0c0  10      -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
856     712     svchost.exe     0xbf0f6b7042c0  20      -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
884     568     fontdrvhost.ex  0xbf0f6b70b1c0  5       -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
892     668     fontdrvhost.ex  0xbf0f6b7091c0  5       -       1       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
976     712     svchost.exe     0xbf0f6b785340  12      -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
320     712     svchost.exe     0xbf0f6b78a2c0  6       -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
564     668     LogonUI.exe     0xbf0f6b7b7100  0       -       1       False   2021-04-30 12:39:44.000000 UTC  2021-04-30 17:39:58.000000 UTC  Disabled
560     668     dwm.exe 0xbf0f6b7ba080  21      -       1       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
1080    712     svchost.exe     0xbf0f6be74340  2       -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
1088    712     svchost.exe     0xbf0f6be77080  3       -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
1164    712     svchost.exe     0xbf0f6becd300  5       -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
1172    712     svchost.exe     0xbf0f6becc080  3       -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
1204    712     svchost.exe     0xbf0f6bed1080  1       -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
1296    712     svchost.exe     0xbf0f6bf192c0  7       -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
1336    712     svchost.exe     0xbf0f6bf20080  8       -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
1376    712     svchost.exe     0xbf0f6be1c080  6       -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
1440    712     svchost.exe     0xbf0f6bf9a0c0  4       -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
1452    712     svchost.exe     0xbf0f6bfa52c0  8       -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
1556    712     svchost.exe     0xbf0f64a81080  8       -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
1660    712     VBoxService.ex  0xbf0f6bff4080  11      -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
1684    712     svchost.exe     0xbf0f6bff5080  5       -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
1784    712     svchost.exe     0xbf0f6c03d340  4       -       0       False   2021-04-30 17:39:46.000000 UTC  N/A     Disabled
1804    712     svchost.exe     0xbf0f6c03b080  5       -       0       False   2021-04-30 17:39:46.000000 UTC  N/A     Disabled
1824    712     svchost.exe     0xbf0f6bfee080  3       -       0       False   2021-04-30 17:39:46.000000 UTC  N/A     Disabled
1880    712     svchost.exe     0xbf0f6c0b1340  2       -       0       False   2021-04-30 17:39:46.000000 UTC  N/A     Disabled
1904    4       MemCompression  0xbf0f6c0b0040  34      -       N/A     False   2021-04-30 17:39:46.000000 UTC  N/A     Disabled
1928    712     svchost.exe     0xbf0f6c0d4340  6       -       0       False   2021-04-30 17:39:46.000000 UTC  N/A     Disabled
1960    712     svchost.exe     0xbf0f6c0d62c0  2       -       0       False   2021-04-30 17:39:46.000000 UTC  N/A     Disabled
1540    712     svchost.exe     0xbf0f6c1af080  3       -       0       False   2021-04-30 17:39:46.000000 UTC  N/A     Disabled
1600    712     svchost.exe     0xbf0f6c1ae300  2       -       0       False   2021-04-30 17:39:46.000000 UTC  N/A     Disabled
1668    712     svchost.exe     0xbf0f6c1b1340  6       -       0       False   2021-04-30 17:39:46.000000 UTC  N/A     Disabled
2168    712     svchost.exe     0xbf0f6c0d3080  9       -       0       False   2021-04-30 17:39:46.000000 UTC  N/A     Disabled
2284    712     svchost.exe     0xbf0f6c280080  11      -       0       False   2021-04-30 17:39:46.000000 UTC  N/A     Disabled
2392    712     svchost.exe     0xbf0f6c2e1080  7       -       0       False   2021-04-30 17:39:46.000000 UTC  N/A     Disabled
2404    712     svchost.exe     0xbf0f6c2b8080  3       -       0       False   2021-04-30 17:39:46.000000 UTC  N/A     Disabled
2420    712     svchost.exe     0xbf0f6c2e6080  6       -       0       False   2021-04-30 17:39:46.000000 UTC  N/A     Disabled
2480    712     svchost.exe     0xbf0f6c2c82c0  8       -       0       False   2021-04-30 17:39:46.000000 UTC  N/A     Disabled
2600    712     spoolsv.exe     0xbf0f6c392280  7       -       0       False   2021-04-30 17:39:46.000000 UTC  N/A     Disabled
2632    712     svchost.exe     0xbf0f6c32a080  13      -       0       False   2021-04-30 17:39:46.000000 UTC  N/A     Disabled
2676    712     svchost.exe     0xbf0f6c39d080  5       -       0       False   2021-04-30 17:39:46.000000 UTC  N/A     Disabled
3008    712     svchost.exe     0xbf0f6c5b02c0  9       -       0       False   2021-04-30 17:39:47.000000 UTC  N/A     Disabled
3016    712     svchost.exe     0xbf0f6c39b080  16      -       0       False   2021-04-30 17:39:47.000000 UTC  N/A     Disabled
3028    712     svchost.exe     0xbf0f6c5d92c0  7       -       0       False   2021-04-30 17:39:47.000000 UTC  N/A     Disabled
3036    712     svchost.exe     0xbf0f6c1342c0  11      -       0       False   2021-04-30 17:39:47.000000 UTC  N/A     Disabled
2764    712     svchost.exe     0xbf0f6c5ee2c0  6       -       0       False   2021-04-30 17:39:47.000000 UTC  N/A     Disabled
2772    712     svchost.exe     0xbf0f6c5f6340  1       -       0       False   2021-04-30 17:39:47.000000 UTC  N/A     Disabled
2840    712     svchost.exe     0xbf0f6bfed080  4       -       0       False   2021-04-30 17:39:47.000000 UTC  N/A     Disabled
                                                                                                                        37.89MB
```

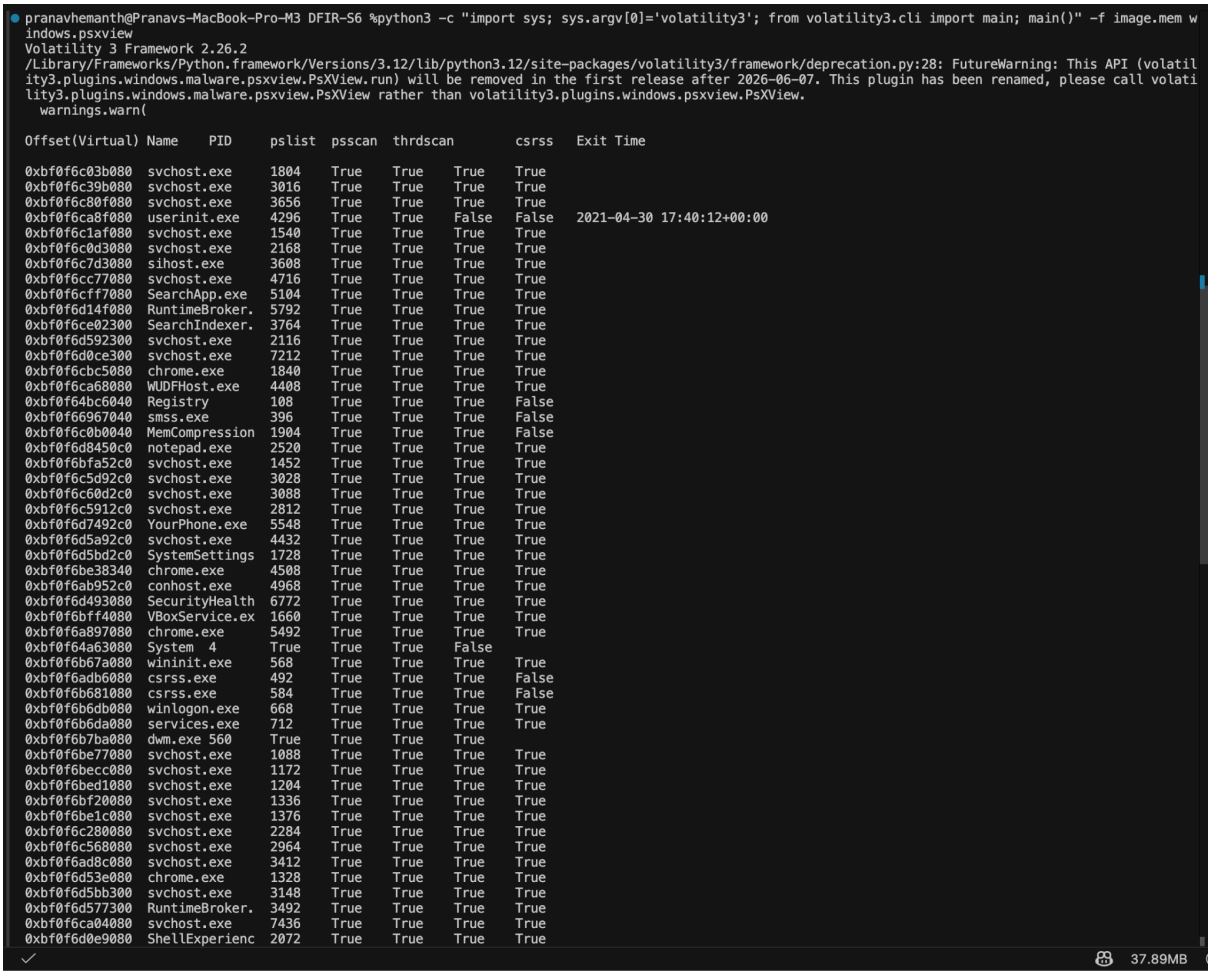
Active running processes were identified using Volatility 3's windows.pslist plugin. This plugin enumerates processes by traversing the kernel's active process list, allowing all processes running at the time the memory image was acquired to be identified.

3. Detect Hidden or Terminated Processes
   Question: How can you detect hidden, terminated, or unlinked processes present in the system memory?
   Answer:
   Command: python3 -c "import sys; sys.argv[0]='volatility3'; from volatility3.cli import main; main()" -f image.mem windows.psscan



```
pranavhemanth@Pranavs-MacBook-Pro-M3 DFIR-S6 %python3 -c "import sys; sys.argv[0]='volatility3'; from volatility3.cli import main; main()" -f image.mem w
indows.psscan
Volatility 3 Framework 2.26.2
Progress:  100.00               PDB scanning finished
PID     PPID    ImageFileName   Offset(V)       Threads Handles SessionId       Wow64   CreateTime      ExitTime        File output

4       0       System  0xbf0f64a63080  132     -       N/A     False   2021-04-30 12:39:40.000000 UTC  N/A     Disabled
1556    712     svchost.exe     0xbf0f64a81080  8       -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
108     4       Registry        0xbf0f64bc6040  4       -       N/A     False   2021-04-30 12:39:38.000000 UTC  N/A     Disabled
4108    1328    chrome.exe      0xbf0f667f1300  12      -       1       False   2021-04-30 17:48:07.000000 UTC  N/A     Disabled
396     4       smss.exe        0xbf0f66967040  2       -       N/A     False   2021-04-30 12:39:40.000000 UTC  N/A     Disabled
6160    2284    audiodg.exe     0xbf0f6a53a080  5       -       0       False   2021-04-30 17:49:52.000000 UTC  N/A     Disabled
3536    5096    FTK Imager.exe  0xbf0f6a707340  20      -       1       True    2021-04-30 17:51:31.000000 UTC  N/A     Disabled
5492    1328    chrome.exe      0xbf0f6a897080  7       -       1       False   2021-04-30 17:44:53.000000 UTC  N/A     Disabled
8160    712     svchost.exe     0xbf0f6a9e2080  4       -       0       False   2021-04-30 17:51:09.000000 UTC  N/A     Disabled
2432    1328    chrome.exe      0xbf0f6a9e7080  19      -       1       False   2021-04-30 17:44:53.000000 UTC  N/A     Disabled
4968    5096    conhost.exe     0xbf0f6ab952c0  5       -       1       False   2021-04-30 17:51:20.000000 UTC  N/A     Disabled
3412    712     svchost.exe     0xbf0f6ad8c080  1       -       0       False   2021-04-30 17:39:47.000000 UTC  N/A     Disabled
492     484     csrss.exe       0xbf0f6adb6080  13      -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
568     484     wininit.exe     0xbf0f6b67a080  1       -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
584     560     csrss.exe       0xbf0f6b681080  16      -       1       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
712     568     services.exe    0xbf0f6b6da080  11      -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
668     560     winlogon.exe    0xbf0f6b6db080  5       -       1       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
736     568     lsass.exe       0xbf0f6b6fb0c0  10      -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
856     712     svchost.exe     0xbf0f6b7042c0  20      -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
892     668     fontdrvhost.ex  0xbf0f6b7091c0  5       -       1       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
884     568     fontdrvhost.ex  0xbf0f6b70b1c0  5       -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
976     712     svchost.exe     0xbf0f6b785340  12      -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
320     712     svchost.exe     0xbf0f6b78a2c0  6       -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
564     668     LogonUI.exe     0xbf0f6b7b7100  0       -       1       False   2021-04-30 12:39:44.000000 UTC  2021-04-30 17:39:58.000000 UTC  Disabled
560     668     dwm.exe 0xbf0f6b7ba080  21      -       1       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
1376    712     svchost.exe     0xbf0f6be1c080  6       -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
4508    1328    chrome.exe      0xbf0f6be38340  10      -       1       False   2021-04-30 17:44:53.000000 UTC  N/A     Disabled
1080    712     svchost.exe     0xbf0f6be74340  2       -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
1088    712     svchost.exe     0xbf0f6be77080  3       -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
1172    712     svchost.exe     0xbf0f6becc080  3       -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
1164    712     svchost.exe     0xbf0f6becd300  5       -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
1204    712     svchost.exe     0xbf0f6bed1080  1       -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
1296    712     svchost.exe     0xbf0f6bf192c0  7       -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
1336    712     svchost.exe     0xbf0f6bf20080  8       -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
1440    712     svchost.exe     0xbf0f6bf9a0c0  4       -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
1452    712     svchost.exe     0xbf0f6bfa52c0  8       -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
2840    712     svchost.exe     0xbf0f6bfed080  4       -       0       False   2021-04-30 17:39:47.000000 UTC  N/A     Disabled
1824    712     svchost.exe     0xbf0f6bfee080  3       -       0       False   2021-04-30 17:39:46.000000 UTC  N/A     Disabled
6696    712     SgrmBroker.exe  0xbf0f6bff1300  7       -       0       False   2021-04-30 17:41:48.000000 UTC  N/A     Disabled
1660    712     VBoxService.ex  0xbf0f6bff4080  11      -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
1684    712     svchost.exe     0xbf0f6bff5080  5       -       0       False   2021-04-30 12:39:44.000000 UTC  N/A     Disabled
1804    712     svchost.exe     0xbf0f6c03b080  5       -       0       False   2021-04-30 17:39:46.000000 UTC  N/A     Disabled
1784    712     svchost.exe     0xbf0f6c03d340  4       -       0       False   2021-04-30 17:39:46.000000 UTC  N/A     Disabled
1904    4       MemCompression  0xbf0f6c0b0040  34      -       N/A     False   2021-04-30 17:39:46.000000 UTC  N/A     Disabled
1880    712     svchost.exe     0xbf0f6c0b1340  2       -       0       False   2021-04-30 17:39:46.000000 UTC  N/A     Disabled
2168    712     svchost.exe     0xbf0f6c0d3080  9       -       0       False   2021-04-30 17:39:46.000000 UTC  N/A     Disabled
1928    712     svchost.exe     0xbf0f6c0d4340  6       -       0       False   2021-04-30 17:39:46.000000 UTC  N/A     Disabled
1960    712     svchost.exe     0xbf0f6c0d62c0  2       -       0       False   2021-04-30 17:39:46.000000 UTC  N/A     Disabled
3036    712     svchost.exe     0xbf0f6c1342c0  11      -       0       False   2021-04-30 17:39:47.000000 UTC  N/A     Disabled
1600    712     svchost.exe     0xbf0f6c1ae300  2       -       0       False   2021-04-30 17:39:46.000000 UTC  N/A     Disabled
1540    712     svchost.exe     0xbf0f6c1af080  3       -       0       False   2021-04-30 17:39:46.000000 UTC  N/A     Disabled
1668    712     svchost.exe     0xbf0f6c1b1340  6       -       0       False   2021-04-30 17:39:46.000000 UTC  N/A     Disabled
2284    712     svchost.exe     0xbf0f6c280080  11      -       0       False   2021-04-30 17:39:46.000000 UTC  N/A     Disabled
2404    712     svchost.exe     0xbf0f6c2b8080  3       -       0       False   2021-04-30 17:39:46.000000 UTC  N/A     Disabled
2480    712     svchost.exe     0xbf0f6c2c82c0  8       -       0       False   2021-04-30 17:39:46.000000 UTC  N/A     Disabled
                                                                                                                        37.89MB
```

Command: python3 -c "import sys; sys.argv[0]='volatility3'; from volatility3.cli import main; main()" -f image.mem windows.psxview



```
pranavhemanth@Pranavs-MacBook-Pro-M3 DFIR-S6 %python3 -c "import sys; sys.argv[0]='volatility3'; from volatility3.cli import main; main()" -f image.mem w
indows.psxview
Volatility 3 Framework 2.26.2
/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages/volatility3/framework/deprecation.py:28: FutureWarning: This API (volatil
ity3.plugins.windows.malware.psxview.PsXView.run) will be removed in the first release after 2026-06-07. This plugin has been renamed, please call volati
lity3.plugins.windows.malware.psxview.PsXView rather than volatility3.plugins.windows.psxview.PsXView.
  warnings.warn(

Offset(Virtual) Name    PID     pslist  psscan  thrdscan        csrss   Exit Time

0xbf0f6c03b080  svchost.exe     1804    True    True    True    True
0xbf0f6c39b080  svchost.exe     3016    True    True    True    True
0xbf0f6c80f080  svchost.exe     3656    True    True    True    True
0xbf0f6ca8f080  userinit.exe    4296    True    True    False   False   2021-04-30 17:40:12+00:00
0xbf0f6c1af080  svchost.exe     1540    True    True    True    True
0xbf0f6c0d3080  svchost.exe     2168    True    True    True    True
0xbf0f6c7d3080  sihost.exe      3608    True    True    True    True
0xbf0f6cc77080  svchost.exe     4716    True    True    True    True
0xbf0f6cff7080  SearchApp.exe   5104    True    True    True    True
0xbf0f6d14f080  RuntimeBroker.  5792    True    True    True    True
0xbf0f6ce02300  SearchIndexer.  3764    True    True    True    True
0xbf0f6d592300  svchost.exe     2116    True    True    True    True
0xbf0f6d0ce300  svchost.exe     7212    True    True    True    True
0xbf0f6cbc5080  chrome.exe      1840    True    True    True    True
0xbf0f6ca68080  WUDFHost.exe    4408    True    True    True    True
0xbf0f64bc6040  Registry        108     True    True    True    False
0xbf0f66967040  smss.exe        396     True    True    True    False
0xbf0f6c0b0040  MemCompression  1904    True    True    True    False
0xbf0f6d8450c0  notepad.exe     2520    True    True    True    True
0xbf0f6bfa52c0  svchost.exe     1452    True    True    True    True
0xbf0f6c5d92c0  svchost.exe     3028    True    True    True    True
0xbf0f6c60d2c0  svchost.exe     3088    True    True    True    True
0xbf0f6c5912c0  svchost.exe     2812    True    True    True    True
0xbf0f6d7492c0  YourPhone.exe   5548    True    True    True    True
0xbf0f6d5a92c0  svchost.exe     4432    True    True    True    True
0xbf0f6d5bd2c0  SystemSettings  1728    True    True    True    True
0xbf0f6be38340  chrome.exe      4508    True    True    True    True
0xbf0f6ab952c0  conhost.exe     4968    True    True    True    True
0xbf0f6d493080  SecurityHealth  6772    True    True    True    True
0xbf0f6bff4080  VBoxService.ex  1660    True    True    True    True
0xbf0f6a897080  chrome.exe      5492    True    True    True    True
0xbf0f64a63080  System  4       True    True    True    False
0xbf0f6b67a080  wininit.exe     568     True    True    True    True
0xbf0f6adb6080  csrss.exe       492     True    True    True    False
0xbf0f6b681080  csrss.exe       584     True    True    True    False
0xbf0f6b6db080  winlogon.exe    668     True    True    True    True
0xbf0f6b6da080  services.exe    712     True    True    True    True
0xbf0f6b7ba080  dwm.exe 560     True    True    True    True
0xbf0f6be77080  svchost.exe     1088    True    True    True    True
0xbf0f6becc080  svchost.exe     1172    True    True    True    True
0xbf0f6bed1080  svchost.exe     1204    True    True    True    True
0xbf0f6bf20080  svchost.exe     1336    True    True    True    True
0xbf0f6be1c080  svchost.exe     1376    True    True    True    True
0xbf0f6c280080  svchost.exe     2284    True    True    True    True
0xbf0f6c568080  svchost.exe     2964    True    True    True    True
0xbf0f6ad8c080  svchost.exe     3412    True    True    True    True
0xbf0f6d53e080  chrome.exe      1328    True    True    True    True
0xbf0f6d5bb300  svchost.exe     3148    True    True    True    True
0xbf0f6d577300  RuntimeBroker.  3492    True    True    True    True
0xbf0f6ca04080  svchost.exe     7436    True    True    True    True
0xbf0f6d0e9080  ShellExperienc  2072    True    True    True    True
```

37.89MB

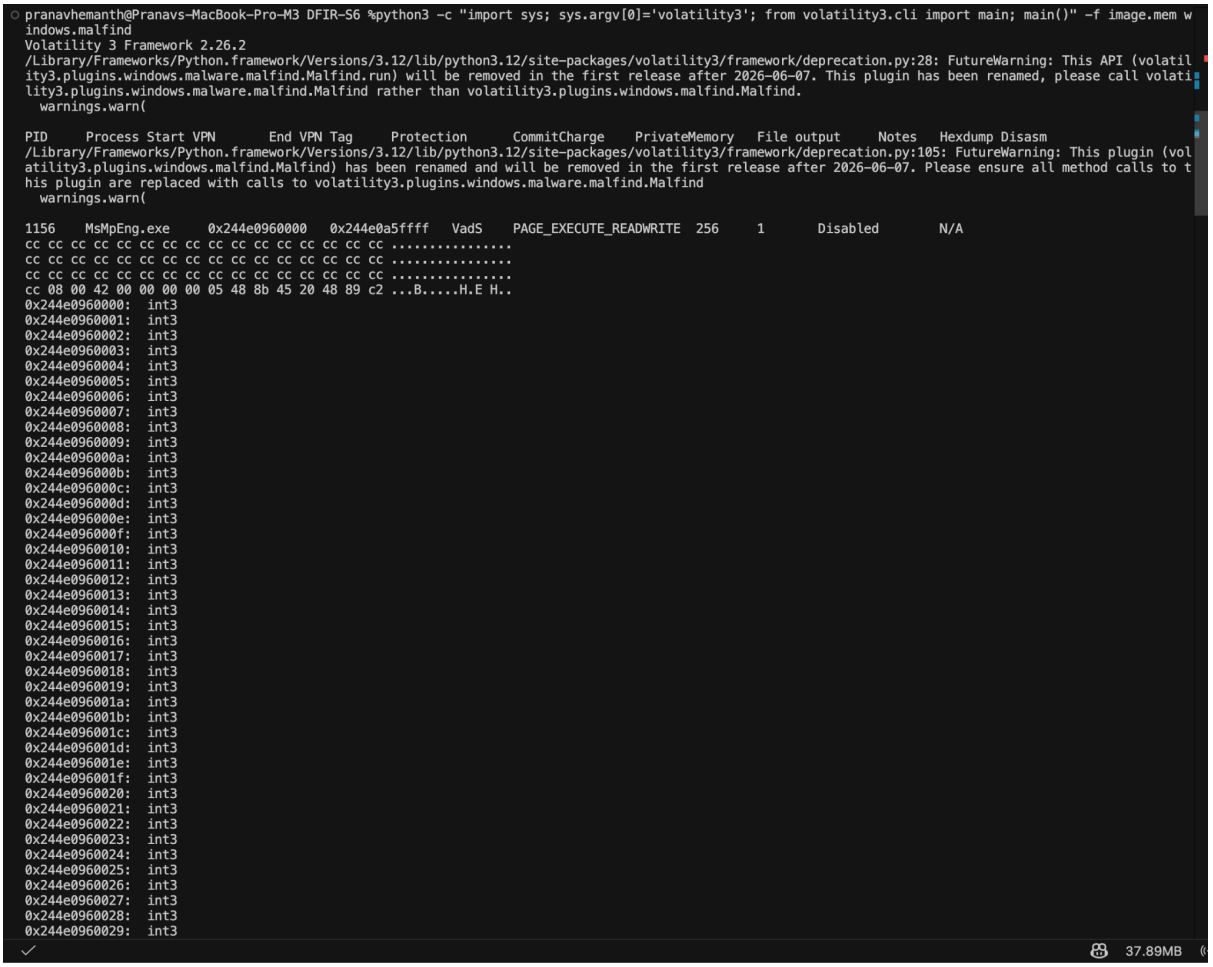
Hidden, terminated, or unlinked processes were identified using Volatility 3's windows.psscan and windows.psxview plugins. The windows.psscan plugin scans memory for residual process objects, allowing detection of terminated or unlinked processes. The windows.psxview plugin compares multiple enumeration techniques to identify discrepancies indicative of hidden processes.

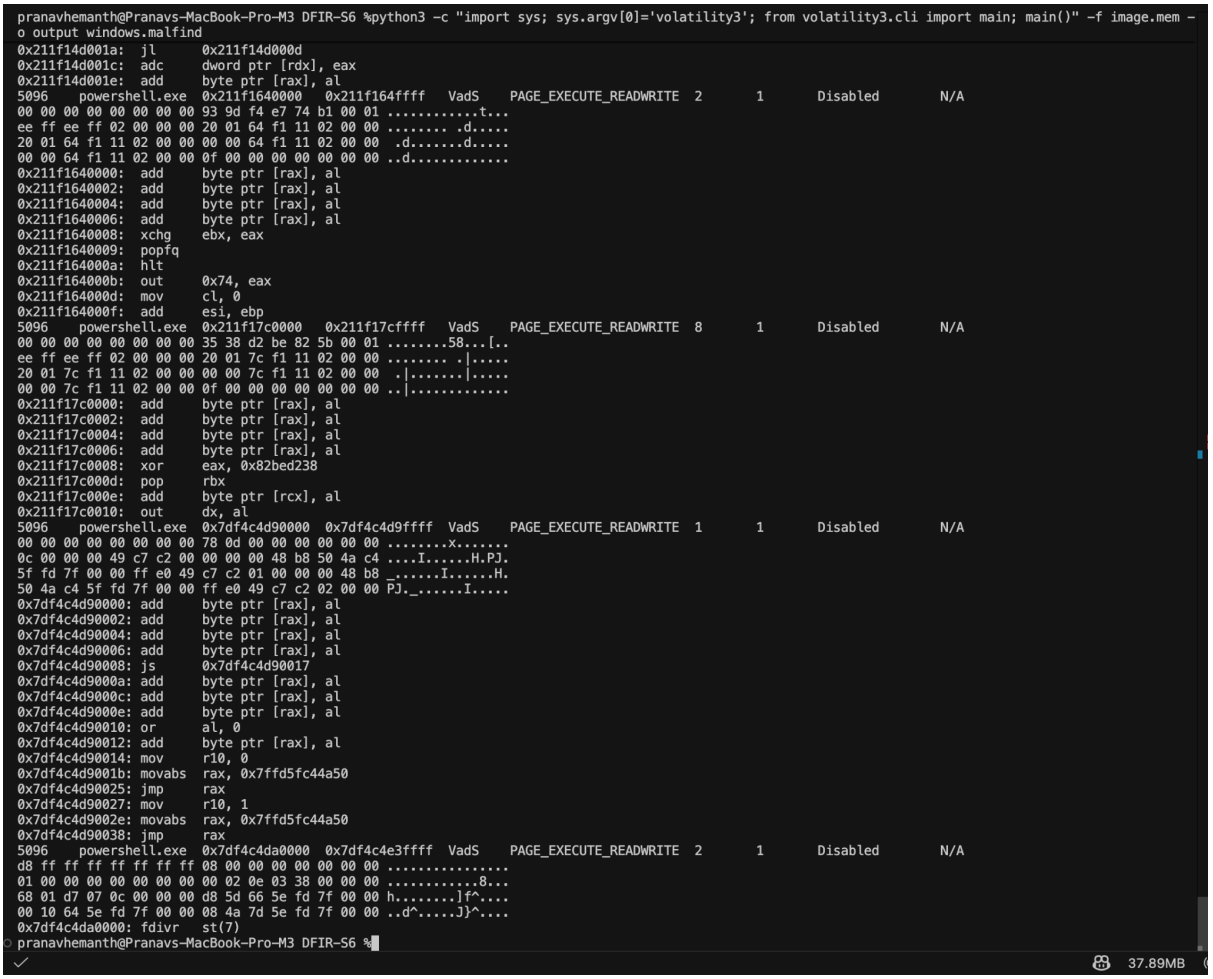
4. Analyze Process Hierarchy
   Question: How can you analyze the parent–child relationship of processes to identify suspicious process spawning?
   Answer:
   Command: python3 -c "import sys; sys.argv[0]='volatility3'; from volatility3.cli import main; main()" -f image.mem windows.pstree

Process hierarchy was analyzed using Volatility 3's windows.pstree plugin. This plugin visualizes parent–child relationships between processes, enabling the identification of anomalous or suspicious process spawning based on deviations from normal process hierarchies.

5. Identify Malicious Code Injection
   Question: How can you detect suspicious or malicious memory regions indicating possible code injection in running processes?
   Answer:
   Command: python3 -c "import sys; sys.argv[0]='volatility3'; from volatility3.cli import main; main()" -f image.mem windows.malfind

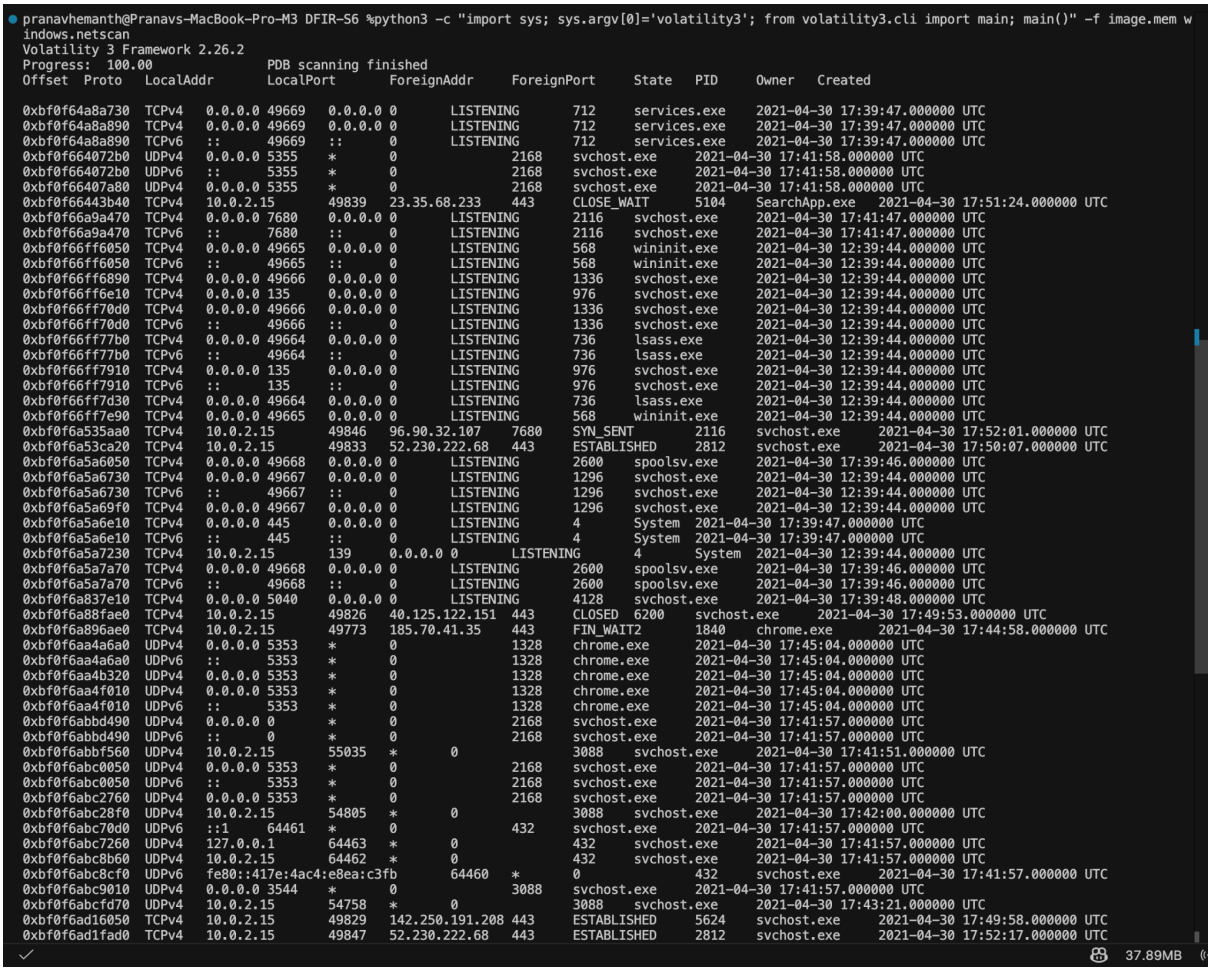
Jan - May 2025 Assignment SUBMISSION_UE23CS342BA6





Suspicious memory regions were detected using Volatility 3's windows.malfind plugin. The identified executable and writable memory regions were dumped to disk for further analysis.

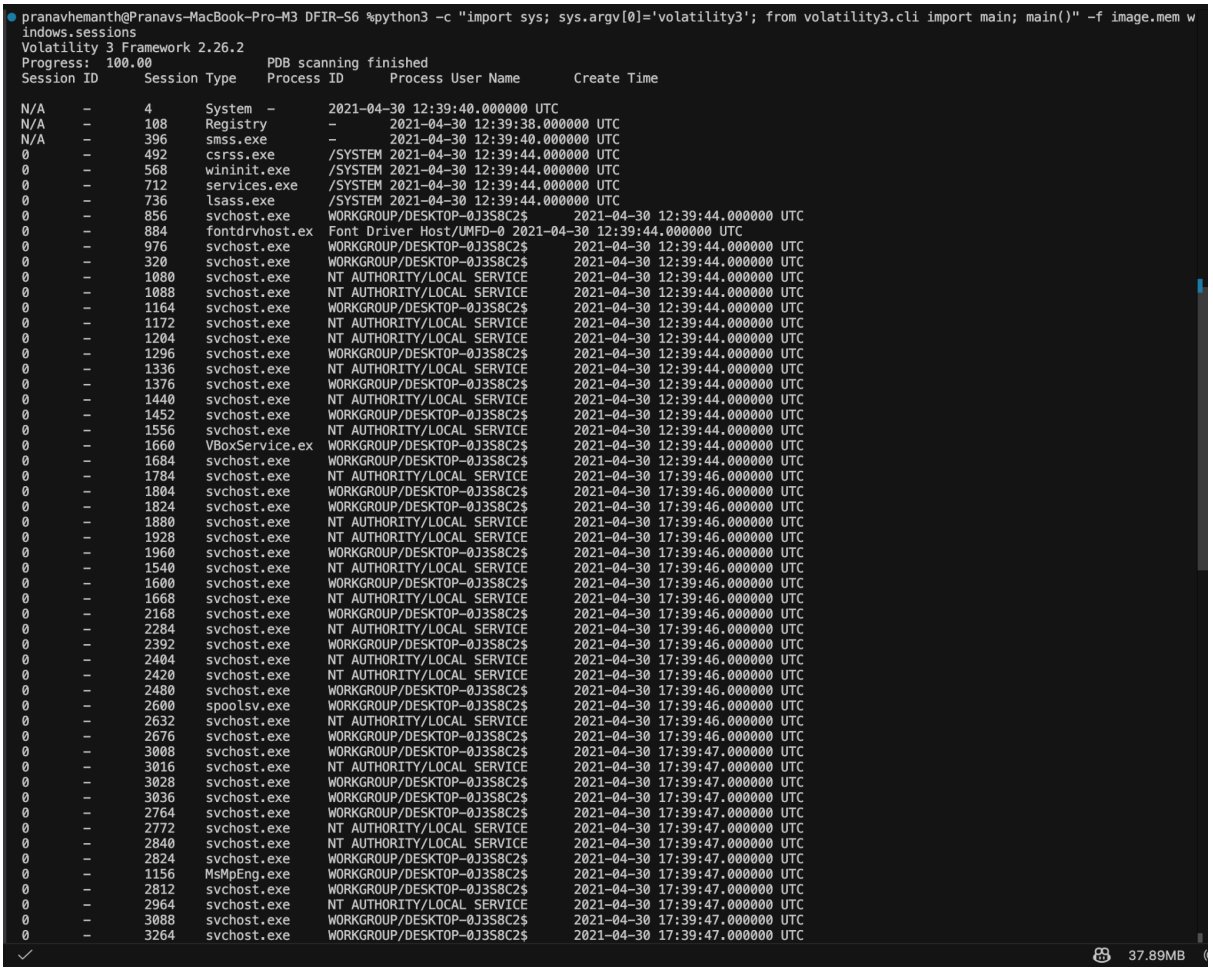
6. Perform Network Connection Analysis
   Question: How can you identify active network connections, listening ports, and remote IP addresses from the memory dump?
   Answer:
   Command: python3 -c "import sys; sys.argv[0]='volatility3'; from volatility3.cli import main; main()" -f image.mem windows.netscan



```
pranavhemanth@Pranavs-MacBook-Pro-M3 DFIR-S6 %python3 -c "import sys; sys.argv[0]='volatility3'; from volatility3.cli import main; main()" -f image.mem w
indows.netscan
Volatility 3 Framework 2.26.2
Progress:  100.00               PDB scanning finished
Offset  Proto   LocalAddr       LocalPort       ForeignAddr     ForeignPort     State   PID     Owner   Created

0xbf0f64a8a730  TCPv4   0.0.0.0 49669   0.0.0.0 0       LISTENING       712     services.exe    2021-04-30 17:39:47.000000 UTC
0xbf0f64a8a890  TCPv4   0.0.0.0 49669   0.0.0.0 0       LISTENING       712     services.exe    2021-04-30 17:39:47.000000 UTC
0xbf0f64a8a890  TCPv6   ::      49669   ::      0       LISTENING       712     services.exe    2021-04-30 17:39:47.000000 UTC
0xbf0f664072b0  UDPv4   0.0.0.0 5355    *       0               2168    svchost.exe     2021-04-30 17:41:58.000000 UTC
0xbf0f664072b0  UDPv6   ::      5355    *       0               2168    svchost.exe     2021-04-30 17:41:58.000000 UTC
0xbf0f66407a80  UDPv4   0.0.0.0 5355    *       0               2168    svchost.exe     2021-04-30 17:41:58.000000 UTC
0xbf0f66443b40  TCPv4   10.0.2.15       49839   23.35.68.233    443     CLOSE_WAIT      5104    SearchApp.exe   2021-04-30 17:51:24.000000 UTC
0xbf0f66a9a470  TCPv4   0.0.0.0 7680    0.0.0.0 0       LISTENING       2116    svchost.exe     2021-04-30 17:41:47.000000 UTC
0xbf0f66a9a470  TCPv6   ::      7680    ::      0       LISTENING       2116    svchost.exe     2021-04-30 17:41:47.000000 UTC
0xbf0f66ff6050  TCPv4   0.0.0.0 49665   0.0.0.0 0       LISTENING       568     wininit.exe     2021-04-30 12:39:44.000000 UTC
0xbf0f66ff6050  TCPv6   ::      49665   ::      0       LISTENING       568     wininit.exe     2021-04-30 12:39:44.000000 UTC
0xbf0f66ff6890  TCPv4   0.0.0.0 49666   0.0.0.0 0       LISTENING       1336    svchost.exe     2021-04-30 12:39:44.000000 UTC
0xbf0f66ff6e10  TCPv4   0.0.0.0 135     0.0.0.0 0       LISTENING       976     svchost.exe     2021-04-30 12:39:44.000000 UTC
0xbf0f66ff70d0  TCPv4   0.0.0.0 49666   0.0.0.0 0       LISTENING       1336    svchost.exe     2021-04-30 12:39:44.000000 UTC
0xbf0f66ff70d0  TCPv6   ::      49666   ::      0       LISTENING       1336    svchost.exe     2021-04-30 12:39:44.000000 UTC
0xbf0f66ff77b0  TCPv4   0.0.0.0 49664   0.0.0.0 0       LISTENING       736     lsass.exe       2021-04-30 12:39:44.000000 UTC
0xbf0f66ff77b0  TCPv6   ::      49664   ::      0       LISTENING       736     lsass.exe       2021-04-30 12:39:44.000000 UTC
0xbf0f66ff7910  TCPv4   0.0.0.0 135     0.0.0.0 0       LISTENING       976     svchost.exe     2021-04-30 12:39:44.000000 UTC
0xbf0f66ff7910  TCPv6   ::      135     ::      0       LISTENING       976     svchost.exe     2021-04-30 12:39:44.000000 UTC
0xbf0f66ff7d30  TCPv4   0.0.0.0 49664   0.0.0.0 0       LISTENING       736     lsass.exe       2021-04-30 12:39:44.000000 UTC
0xbf0f66ff7e90  TCPv4   0.0.0.0 49665   0.0.0.0 0       LISTENING       568     wininit.exe     2021-04-30 12:39:44.000000 UTC
0xbf0f6a535aa0  TCPv4   10.0.2.15       49846   96.90.32.107    7680    SYN_SENT        2116    svchost.exe     2021-04-30 17:52:01.000000 UTC
0xbf0f6a53ca20  TCPv4   10.0.2.15       49833   52.230.222.68   443     ESTABLISHED     2812    svchost.exe     2021-04-30 17:50:07.000000 UTC
0xbf0f6a5a6050  TCPv4   0.0.0.0 49668   0.0.0.0 0       LISTENING       2600    spoolsv.exe     2021-04-30 17:39:46.000000 UTC
0xbf0f6a5a6730  TCPv4   0.0.0.0 49667   0.0.0.0 0       LISTENING       1296    svchost.exe     2021-04-30 12:39:44.000000 UTC
0xbf0f6a5a6730  TCPv6   ::      49667   ::      0       LISTENING       1296    svchost.exe     2021-04-30 12:39:44.000000 UTC
0xbf0f6a5a69f0  TCPv4   0.0.0.0 49667   0.0.0.0 0       LISTENING       1296    svchost.exe     2021-04-30 12:39:44.000000 UTC
0xbf0f6a5a6e10  TCPv4   0.0.0.0 445     0.0.0.0 0       LISTENING       4       System  2021-04-30 17:39:47.000000 UTC
0xbf0f6a5a6e10  TCPv6   ::      445     ::      0       LISTENING       4       System  2021-04-30 17:39:47.000000 UTC
0xbf0f6a5a7230  TCPv4   10.0.2.15       139     0.0.0.0 0       LISTENING       4       System  2021-04-30 12:39:44.000000 UTC
0xbf0f6a5a7a70  TCPv4   0.0.0.0 49668   0.0.0.0 0       LISTENING       2600    spoolsv.exe     2021-04-30 17:39:46.000000 UTC
0xbf0f6a5a7a70  TCPv6   ::      49668   ::      0       LISTENING       2600    spoolsv.exe     2021-04-30 17:39:46.000000 UTC
0xbf0f6a837e10  TCPv4   0.0.0.0 5040    0.0.0.0 0       LISTENING       4128    svchost.exe     2021-04-30 17:39:48.000000 UTC
0xbf0f6a88fae0  TCPv4   10.0.2.15       49826   40.125.122.151  443     CLOSED  6200    svchost.exe     2021-04-30 17:49:53.000000 UTC
0xbf0f6a896ae0  TCPv4   10.0.2.15       49773   185.70.41.35    443     FIN_WAIT2       1840    chrome.exe      2021-04-30 17:44:58.000000 UTC
0xbf0f6aa4a6a0  UDPv4   0.0.0.0 5353    *       0               1328    chrome.exe      2021-04-30 17:45:04.000000 UTC
0xbf0f6aa4a6a0  UDPv6   ::      5353    *       0               1328    chrome.exe      2021-04-30 17:45:04.000000 UTC
0xbf0f6aa4b320  UDPv4   0.0.0.0 5353    *       0               1328    chrome.exe      2021-04-30 17:45:04.000000 UTC
0xbf0f6aa4f010  UDPv4   0.0.0.0 5353    *       0               1328    chrome.exe      2021-04-30 17:45:04.000000 UTC
0xbf0f6aa4f010  UDPv6   ::      5353    *       0               1328    chrome.exe      2021-04-30 17:45:04.000000 UTC
0xbf0f6abbd490  UDPv4   0.0.0.0 0       *       0               2168    svchost.exe     2021-04-30 17:41:57.000000 UTC
0xbf0f6abbd490  UDPv6   ::      0       *       0               2168    svchost.exe     2021-04-30 17:41:57.000000 UTC
0xbf0f6abbf560  UDPv4   10.0.2.15       55035   *       0               3088    svchost.exe     2021-04-30 17:41:51.000000 UTC
0xbf0f6abc0050  UDPv4   0.0.0.0 5353    *       0               2168    svchost.exe     2021-04-30 17:41:57.000000 UTC
0xbf0f6abc0050  UDPv6   ::      5353    *       0               2168    svchost.exe     2021-04-30 17:41:57.000000 UTC
0xbf0f6abc2760  UDPv4   0.0.0.0 5353    *       0               2168    svchost.exe     2021-04-30 17:41:57.000000 UTC
0xbf0f6abc28f0  UDPv4   10.0.2.15       54805   *       0               3088    svchost.exe     2021-04-30 17:42:00.000000 UTC
0xbf0f6abc70d0  UDPv6   ::1     64461   *       0               432     svchost.exe     2021-04-30 17:41:57.000000 UTC
0xbf0f6abc7260  UDPv4   127.0.0.1       64463   *       0               432     svchost.exe     2021-04-30 17:41:57.000000 UTC
0xbf0f6abc8b60  UDPv4   10.0.2.15       64462   *       0               432     svchost.exe     2021-04-30 17:41:57.000000 UTC
0xbf0f6abc8cf0  UDPv6   fe80::417e:4ac4:e8ea:c3fb       64460   *       0               432     svchost.exe     2021-04-30 17:41:57.000000 UTC
0xbf0f6abc9010  UDPv4   0.0.0.0 3544    *       0               3088    svchost.exe     2021-04-30 17:41:57.000000 UTC
0xbf0f6abcfd70  UDPv4   10.0.2.15       54758   *       0               3088    svchost.exe     2021-04-30 17:43:21.000000 UTC
0xbf0f6ad16050  TCPv4   10.0.2.15       49829   142.250.191.208 443     ESTABLISHED     5624    svchost.exe     2021-04-30 17:49:58.000000 UTC
0xbf0f6ad1fad0  TCPv4   10.0.2.15       49847   52.230.222.68   443     ESTABLISHED     2812    svchost.exe     2021-04-30 17:52:17.000000 UTC
```


Network activity was analyzed using Volatility 3's windows.netscan plugin. This plugin scans memory for network socket structures and identifies active and listening connections, including local and remote IP addresses, ports, connection states, and the associated processes.

7. Analyze User Sessions
   Question: How can you identify active and disconnected user sessions present in the memory image?
   Answer:
   Command: python3 -c "import sys; sys.argv[0]='volatility3'; from volatility3.cli import main; main()" -f image.mem windows.sessions



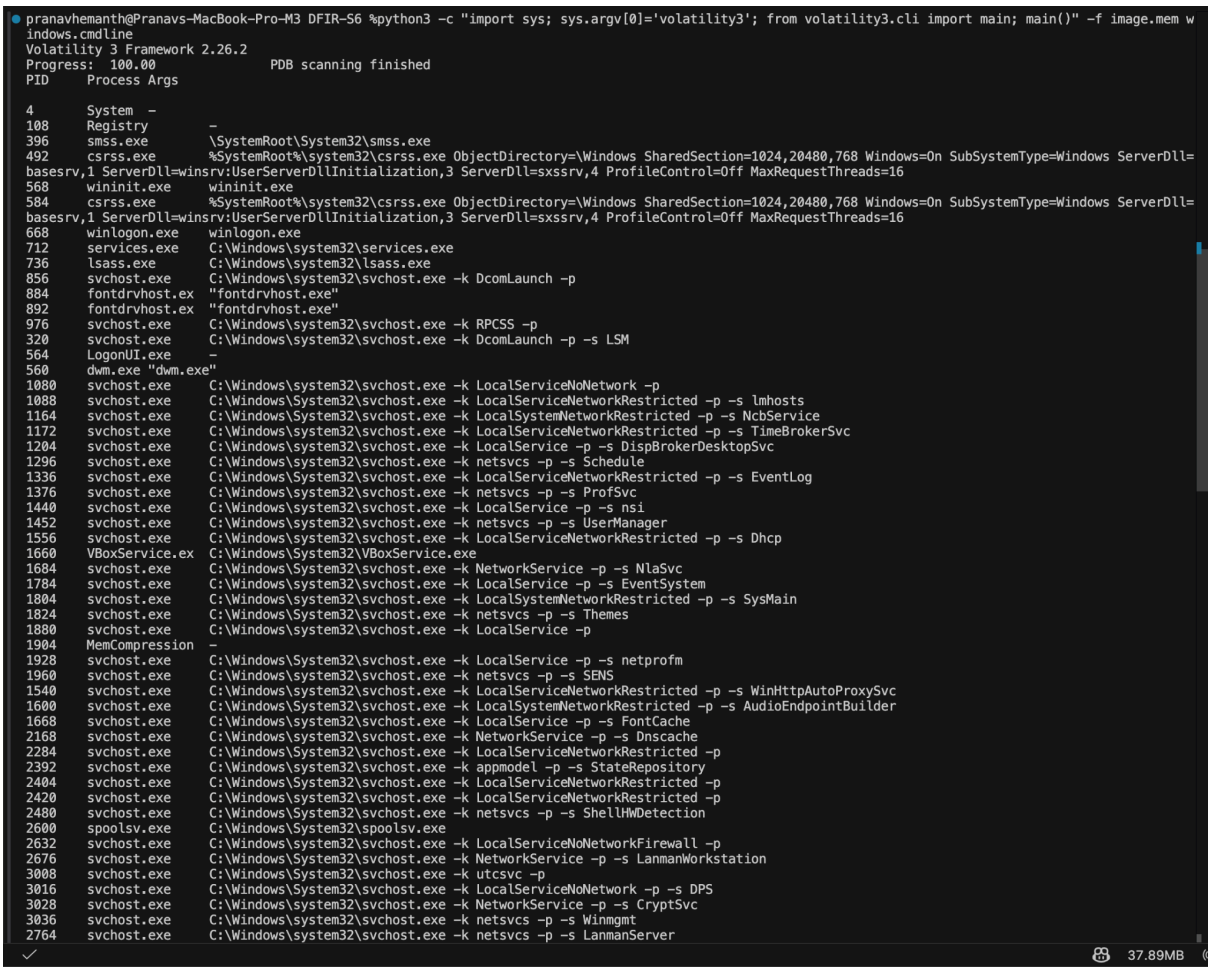
```
pranavhemanth@Pranavs-MacBook-Pro-M3 DFIR-S6 %python3 -c "import sys; sys.argv[0]='volatility3'; from volatility3.cli import main; main()" -f image.mem windows.sessions
Volatility 3 Framework 2.26.2
Progress:  100.00               PDB scanning finished
Session ID      Session Type    Process ID      Process User Name       Create Time

N/A     -       4       System  -       2021-04-30 12:39:40.000000 UTC
N/A     -       108     Registry        -       2021-04-30 12:39:38.000000 UTC
N/A     -       396     smss.exe        -       2021-04-30 12:39:40.000000 UTC
0       -       492     csrss.exe       /SYSTEM 2021-04-30 12:39:44.000000 UTC
0       -       568     wininit.exe     /SYSTEM 2021-04-30 12:39:44.000000 UTC
0       -       712     services.exe    /SYSTEM 2021-04-30 12:39:44.000000 UTC
0       -       736     lsass.exe       /SYSTEM 2021-04-30 12:39:44.000000 UTC
0       -       856     svchost.exe     WORKGROUP/DESKTOP-0J3S8C2$      2021-04-30 12:39:44.000000 UTC
0       -       884     fontdrvhost.ex  Font Driver Host/UMFD-0 2021-04-30 12:39:44.000000 UTC
0       -       976     svchost.exe     WORKGROUP/DESKTOP-0J3S8C2$      2021-04-30 12:39:44.000000 UTC
0       -       320     svchost.exe     WORKGROUP/DESKTOP-0J3S8C2$      2021-04-30 12:39:44.000000 UTC
0       -       1080    svchost.exe     NT AUTHORITY/LOCAL SERVICE      2021-04-30 12:39:44.000000 UTC
0       -       1088    svchost.exe     NT AUTHORITY/LOCAL SERVICE      2021-04-30 12:39:44.000000 UTC
0       -       1164    svchost.exe     WORKGROUP/DESKTOP-0J3S8C2$      2021-04-30 12:39:44.000000 UTC
0       -       1172    svchost.exe     NT AUTHORITY/LOCAL SERVICE      2021-04-30 12:39:44.000000 UTC
0       -       1204    svchost.exe     NT AUTHORITY/LOCAL SERVICE      2021-04-30 12:39:44.000000 UTC
0       -       1296    svchost.exe     WORKGROUP/DESKTOP-0J3S8C2$      2021-04-30 12:39:44.000000 UTC
0       -       1336    svchost.exe     NT AUTHORITY/LOCAL SERVICE      2021-04-30 12:39:44.000000 UTC
0       -       1376    svchost.exe     WORKGROUP/DESKTOP-0J3S8C2$      2021-04-30 12:39:44.000000 UTC
0       -       1440    svchost.exe     NT AUTHORITY/LOCAL SERVICE      2021-04-30 12:39:44.000000 UTC
0       -       1452    svchost.exe     WORKGROUP/DESKTOP-0J3S8C2$      2021-04-30 12:39:44.000000 UTC
0       -       1556    svchost.exe     NT AUTHORITY/LOCAL SERVICE      2021-04-30 12:39:44.000000 UTC
0       -       1660    VBoxService.ex  WORKGROUP/DESKTOP-0J3S8C2$      2021-04-30 12:39:44.000000 UTC
0       -       1684    svchost.exe     WORKGROUP/DESKTOP-0J3S8C2$      2021-04-30 12:39:44.000000 UTC
0       -       1784    svchost.exe     NT AUTHORITY/LOCAL SERVICE      2021-04-30 17:39:46.000000 UTC
0       -       1804    svchost.exe     WORKGROUP/DESKTOP-0J3S8C2$      2021-04-30 17:39:46.000000 UTC
0       -       1824    svchost.exe     WORKGROUP/DESKTOP-0J3S8C2$      2021-04-30 17:39:46.000000 UTC
0       -       1880    svchost.exe     NT AUTHORITY/LOCAL SERVICE      2021-04-30 17:39:46.000000 UTC
0       -       1928    svchost.exe     NT AUTHORITY/LOCAL SERVICE      2021-04-30 17:39:46.000000 UTC
0       -       1960    svchost.exe     WORKGROUP/DESKTOP-0J3S8C2$      2021-04-30 17:39:46.000000 UTC
0       -       1540    svchost.exe     NT AUTHORITY/LOCAL SERVICE      2021-04-30 17:39:46.000000 UTC
0       -       1600    svchost.exe     WORKGROUP/DESKTOP-0J3S8C2$      2021-04-30 17:39:46.000000 UTC
0       -       1668    svchost.exe     NT AUTHORITY/LOCAL SERVICE      2021-04-30 17:39:46.000000 UTC
0       -       2168    svchost.exe     WORKGROUP/DESKTOP-0J3S8C2$      2021-04-30 17:39:46.000000 UTC
0       -       2284    svchost.exe     NT AUTHORITY/LOCAL SERVICE      2021-04-30 17:39:46.000000 UTC
0       -       2392    svchost.exe     WORKGROUP/DESKTOP-0J3S8C2$      2021-04-30 17:39:46.000000 UTC
0       -       2404    svchost.exe     NT AUTHORITY/LOCAL SERVICE      2021-04-30 17:39:46.000000 UTC
0       -       2420    svchost.exe     NT AUTHORITY/LOCAL SERVICE      2021-04-30 17:39:46.000000 UTC
0       -       2480    svchost.exe     WORKGROUP/DESKTOP-0J3S8C2$      2021-04-30 17:39:46.000000 UTC
0       -       2600    spoolsv.exe     WORKGROUP/DESKTOP-0J3S8C2$      2021-04-30 17:39:46.000000 UTC
0       -       2632    svchost.exe     NT AUTHORITY/LOCAL SERVICE      2021-04-30 17:39:46.000000 UTC
0       -       2676    svchost.exe     WORKGROUP/DESKTOP-0J3S8C2$      2021-04-30 17:39:46.000000 UTC
0       -       3008    svchost.exe     WORKGROUP/DESKTOP-0J3S8C2$      2021-04-30 17:39:47.000000 UTC
0       -       3016    svchost.exe     NT AUTHORITY/LOCAL SERVICE      2021-04-30 17:39:47.000000 UTC
0       -       3028    svchost.exe     WORKGROUP/DESKTOP-0J3S8C2$      2021-04-30 17:39:47.000000 UTC
0       -       3036    svchost.exe     WORKGROUP/DESKTOP-0J3S8C2$      2021-04-30 17:39:47.000000 UTC
0       -       2764    svchost.exe     WORKGROUP/DESKTOP-0J3S8C2$      2021-04-30 17:39:47.000000 UTC
0       -       2772    svchost.exe     NT AUTHORITY/LOCAL SERVICE      2021-04-30 17:39:47.000000 UTC
0       -       2840    svchost.exe     NT AUTHORITY/LOCAL SERVICE      2021-04-30 17:39:47.000000 UTC
0       -       2824    svchost.exe     WORKGROUP/DESKTOP-0J3S8C2$      2021-04-30 17:39:47.000000 UTC
0       -       1156    MsMpEng.exe     WORKGROUP/DESKTOP-0J3S8C2$      2021-04-30 17:39:47.000000 UTC
0       -       2812    svchost.exe     WORKGROUP/DESKTOP-0J3S8C2$      2021-04-30 17:39:47.000000 UTC
0       -       2964    svchost.exe     NT AUTHORITY/LOCAL SERVICE      2021-04-30 17:39:47.000000 UTC
0       -       3088    svchost.exe     WORKGROUP/DESKTOP-0J3S8C2$      2021-04-30 17:39:47.000000 UTC
0       -       3264    svchost.exe     WORKGROUP/DESKTOP-0J3S8C2$      2021-04-30 17:39:47.000000 UTC
```

37.89MB

User sessions were analyzed using Volatility 3's windows.sessions plugin. This plugin identifies active and disconnected user logon sessions by analyzing session-related kernel data structures present in the memory image.

8. Extract Command Execution History
   Question: How can you extract command-line arguments and console command history executed by users from memory?
   Answer:
   Command: python3 -c "import sys; sys.argv[0]='volatility3'; from volatility3.cli import main; main()" -f image.mem windows.cmdline

```
pranavhemanth@Pranavs-MacBook-Pro-M3 DFIR-S6 %python3 -c "import sys; sys.argv[0]='volatility3'; from volatility3.cli import main; main()" -f image.mem w
indows.cmdline
Volatility 3 Framework 2.26.2
Progress:  100.00               PDB scanning finished
PID     Process Args

4       System  -
108     Registry        -
396     smss.exe        \SystemRoot\System32\smss.exe
492     csrss.exe       %SystemRoot%\system32\csrss.exe ObjectDirectory=\Windows SharedSection=1024,20480,768 Windows=On SubSystemType=Windows ServerDll=
basesrv,1 ServerDll=winsrv:UserServerDllInitialization,3 ServerDll=sxssrv,4 ProfileControl=Off MaxRequestThreads=16
568     wininit.exe     wininit.exe
584     csrss.exe       %SystemRoot%\system32\csrss.exe ObjectDirectory=\Windows SharedSection=1024,20480,768 Windows=On SubSystemType=Windows ServerDll=
basesrv,1 ServerDll=winsrv:UserServerDllInitialization,3 ServerDll=sxssrv,4 ProfileControl=Off MaxRequestThreads=16
668     winlogon.exe    winlogon.exe
712     services.exe    C:\Windows\system32\services.exe
736     lsass.exe       C:\Windows\system32\lsass.exe
856     svchost.exe     C:\Windows\system32\svchost.exe -k DcomLaunch -p
884     fontdrvhost.ex  "fontdrvhost.exe"
892     fontdrvhost.ex  "fontdrvhost.exe"
976     svchost.exe     C:\Windows\system32\svchost.exe -k RPCSS -p
320     svchost.exe     C:\Windows\system32\svchost.exe -k DcomLaunch -p -s LSM
564     LogonUI.exe     -
560     dwm.exe "dwm.exe"
1080    svchost.exe     C:\Windows\system32\svchost.exe -k LocalServiceNoNetwork -p
1088    svchost.exe     C:\Windows\System32\svchost.exe -k LocalServiceNetworkRestricted -p -s lmhosts
1164    svchost.exe     C:\Windows\System32\svchost.exe -k LocalSystemNetworkRestricted -p -s NcbService
1172    svchost.exe     C:\Windows\system32\svchost.exe -k LocalServiceNetworkRestricted -p -s TimeBrokerSvc
1204    svchost.exe     C:\Windows\system32\svchost.exe -k LocalService -p -s DispBrokerDesktopSvc
1296    svchost.exe     C:\Windows\system32\svchost.exe -k netsvcs -p -s Schedule
1336    svchost.exe     C:\Windows\System32\svchost.exe -k LocalServiceNetworkRestricted -p -s EventLog
1376    svchost.exe     C:\Windows\system32\svchost.exe -k netsvcs -p -s ProfSvc
1440    svchost.exe     C:\Windows\system32\svchost.exe -k LocalService -p -s nsi
1452    svchost.exe     C:\Windows\system32\svchost.exe -k netsvcs -p -s UserManager
1556    svchost.exe     C:\Windows\system32\svchost.exe -k LocalServiceNetworkRestricted -p -s Dhcp
1660    VBoxService.ex  C:\Windows\System32\VBoxService.exe
1684    svchost.exe     C:\Windows\System32\svchost.exe -k NetworkService -p -s NlaSvc
1784    svchost.exe     C:\Windows\system32\svchost.exe -k LocalService -p -s EventSystem
1804    svchost.exe     C:\Windows\system32\svchost.exe -k LocalSystemNetworkRestricted -p -s SysMain
1824    svchost.exe     C:\Windows\System32\svchost.exe -k netsvcs -p -s Themes
1880    svchost.exe     C:\Windows\system32\svchost.exe -k LocalService -p
1904    MemCompression  -
1928    svchost.exe     C:\Windows\System32\svchost.exe -k LocalService -p -s netprofm
1960    svchost.exe     C:\Windows\system32\svchost.exe -k netsvcs -p -s SENS
1540    svchost.exe     C:\Windows\system32\svchost.exe -k LocalServiceNetworkRestricted -p -s WinHttpAutoProxySvc
1600    svchost.exe     C:\Windows\System32\svchost.exe -k LocalSystemNetworkRestricted -p -s AudioEndpointBuilder
1668    svchost.exe     C:\Windows\system32\svchost.exe -k LocalService -p -s FontCache
2168    svchost.exe     C:\Windows\system32\svchost.exe -k NetworkService -p -s Dnscache
2284    svchost.exe     C:\Windows\System32\svchost.exe -k LocalServiceNetworkRestricted -p
2392    svchost.exe     C:\Windows\system32\svchost.exe -k appmodel -p -s StateRepository
2404    svchost.exe     C:\Windows\System32\svchost.exe -k LocalServiceNetworkRestricted -p
2420    svchost.exe     C:\Windows\System32\svchost.exe -k LocalServiceNetworkRestricted -p
2480    svchost.exe     C:\Windows\System32\svchost.exe -k netsvcs -p -s ShellHWDetection
2600    spoolsv.exe     C:\Windows\System32\spoolsv.exe
2632    svchost.exe     C:\Windows\System32\svchost.exe -k LocalServiceNoNetworkFirewall -p
2676    svchost.exe     C:\Windows\System32\svchost.exe -k NetworkService -p -s LanmanWorkstation
3008    svchost.exe     C:\Windows\System32\svchost.exe -k utcsvc -p
3016    svchost.exe     C:\Windows\System32\svchost.exe -k LocalServiceNoNetwork -p -s DPS
3028    svchost.exe     C:\Windows\system32\svchost.exe -k NetworkService -p -s CryptSvc
3036    svchost.exe     C:\Windows\system32\svchost.exe -k netsvcs -p -s Winmgmt
2764    svchost.exe     C:\Windows\system32\svchost.exe -k netsvcs -p -s LanmanServer
```

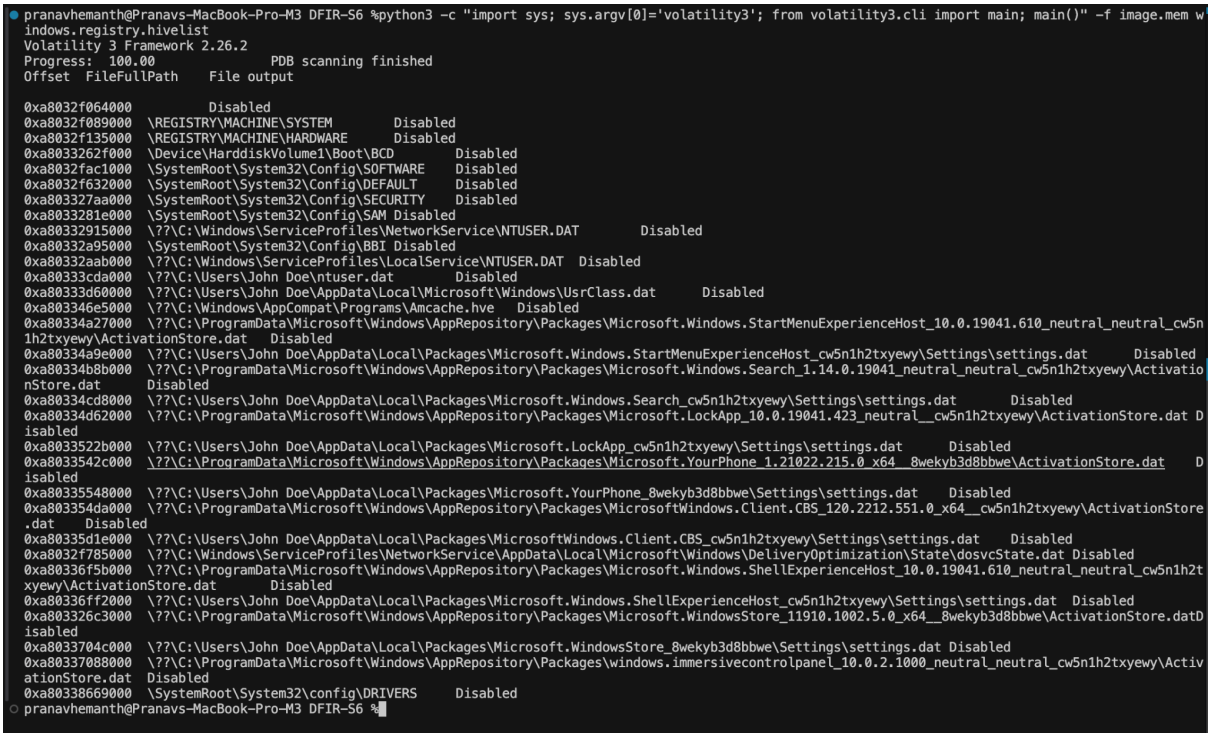


Command: python3 -c "import sys; sys.argv[0]='volatility3'; from volatility3.cli import main; main()" -f image.mem windows.consoles

```
pranavhemanth@Pranavs-MacBook-Pro-M3 DFIR-S6 %python3 -c "import sys; sys.argv[0]='volatility3'; from volatility3.cli import main; main()" -f image.mem w
indows.consoles
Volatility 3 Framework 2.26.2
Progress:  100.00               PDB scanning finished
PID     Process ConsoleInfo     Property        Address Data

4968    conhost.exe     0x7ff72e77e9d0  _CONSOLE_INFORMATION    0x7ff72e77e9d0  -
* 4968  conhost.exe     0x7ff72e77e9d0  _CONSOLE_INFORMATION.ScreenX    0x7ff72e77e9f0  120
* 4968  conhost.exe     0x7ff72e77e9d0  _CONSOLE_INFORMATION.ScreenY    0x7ff72e77e9f2  3000
* 4968  conhost.exe     0x7ff72e77e9d0  _CONSOLE_INFORMATION.CommandHistorySize 0x7ff72e77ea60  50
* 4968  conhost.exe     0x7ff72e77e9d0  _CONSOLE_INFORMATION.HistoryBufferCount 0x7ff72e77e878  1
* 4968  conhost.exe     0x7ff72e77e9d0  _CONSOLE_INFORMATION.HistoryBufferMax   0x7ff72e77ea64  4
* 4968  conhost.exe     0x7ff72e77e9d0  _CONSOLE_INFORMATION.Title      0x7ff72e77f068  Administrator: Windows PowerShell
* 4968  conhost.exe     0x7ff72e77e9d0  _CONSOLE_INFORMATION.OriginalTitle      0x7ff72e77f0a8  Windows PowerShell
* 4968  conhost.exe     0x7ff72e77e9d0  _CONSOLE_INFORMATION.ProcessCount       0x7ff72e77ef30  1
* 4968  conhost.exe     0x7ff72e77e9d0  _CONSOLE_INFORMATION.ConsoleProcessList 0x7ff72e77ef28  -
** 4968 conhost.exe     0x7ff72e77e9d0  _CONSOLE_INFORMATION.ConsoleProcessList.ConsoleProcess_0        0x1f805206cd0   -
** 4968 conhost.exe     0x7ff72e77e9d0  _CONSOLE_INFORMATION.ConsoleProcessList.ConsoleProcess_0_ProcessId      0x1f805206cec   5096
** 4968 conhost.exe     0x7ff72e77e9d0  _CONSOLE_INFORMATION.ConsoleProcessList.ConsoleProcess_0_ProcessHandle  0x1f805206d00   0x100
* 4968  conhost.exe     0x7ff72e77e9d0  _CONSOLE_INFORMATION.ExeAliasList       0x7ff72e77ef6c  -
* 4968  conhost.exe     0x7ff72e77e9d0  _CONSOLE_INFORMATION.HistoryList        0x7ff72e77e870  -
** 4968 conhost.exe     0x7ff72e77e9d0  _CONSOLE_INFORMATION.HistoryList.CommandHistory_0       0x1f8072315a0   -
** 4968 conhost.exe     0x7ff72e77e9d0  _CONSOLE_INFORMATION.HistoryList.CommandHistory_0_Application    0x1f8072315d0   powershell.exe
** 4968 conhost.exe     0x7ff72e77e9d0  _CONSOLE_INFORMATION.HistoryList.CommandHistory_0_ProcessHandle 0x1f805206d00   0x100
** 4968 conhost.exe     0x7ff72e77e9d0  _CONSOLE_INFORMATION.HistoryList.CommandHistory_0_CommandCount  N/A     -
** 4968 conhost.exe     0x7ff72e77e9d0  _CONSOLE_INFORMATION.HistoryList.CommandHistory_0_LastDisplayed 0x1f8072315fc   -1
* 4968  conhost.exe     0x7ff72e77e9d0  _CONSOLE_INFORMATION.CurrentScreenBuffer        0x7ff72e77ef40  -
** 4968 conhost.exe     0x7ff72e77e9d0  _CONSOLE_INFORMATION.ScreenBuffer_0     0x1f80523e700   -
** 4968 conhost.exe     0x7ff72e77e9d0  _CONSOLE_INFORMATION.ScreenBuffer_0.ScreenX     N/A     120
** 4968 conhost.exe     0x7ff72e77e9d0  _CONSOLE_INFORMATION.ScreenBuffer_0.ScreenY     N/A     3000
** 4968 conhost.exe     0x7ff72e77e9d0  _CONSOLE_INFORMATION.ScreenBuffer_0.Dump        N/A     Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Windows\system32> cd E:\FTK_Imager_Lite_3.1.1\
PS E:\FTK_Imager_Lite_3.1.1> & '.\FTK Imager.exe'
PS E:\FTK_Imager_Lite_3.1.1>
pranavhemanth@Pranavs-MacBook-Pro-M3 DFIR-S6 %
```

Command execution history was extracted using Volatility 3's windows.cmdline and windows.consoles plugins. The windows.cmdline plugin retrieves command-line arguments used to start processes, while the windows.consoles plugin extracts interactive command history from Windows console buffers present in memory.

9. Extract Registry Hives from Memory
   Question: How can you identify and extract Windows registry hives stored in system memory?
   Answer:
   Command: python3 -c "import sys; sys.argv[0]='volatility3'; from volatility3.cli import main; main()" -f image.mem windows.registry.hivelist



Command: python3 -c "import sys; sys.argv[0]='volatility3'; from volatility3.cli import main; main()" -f image.mem windows.registry.hivescan

```
pranavhemanth@Pranavs-MacBook-Pro-M3 DFIR-S6 %python3 -c "import sys; sys.argv[0]='volatility3'; from volatility3.cli import main; main()" -f image.mem w
indows.registry.hivescan
Volatility 3 Framework 2.26.2
Progress:  100.00               PDB scanning finished
Offset

0xa80332a95000
0xa80332aab000
0xa80334cd8000
0xa80334b8b000
0xa8032f089000
0xa8032f064000
0xa8033281e000
0xa80336ff2000
0xa8033542c000
0xa80336f5b000
0xa8032f785000
0xa803354da000
0xa8033522b000
0xa80333d60000
0xa80334a27000
0xa80335548000
0xa8033704c000
0xa80335d1e000
0xa80338669000
0xa8033262f000
0xa8032fac1000
0xa803327aa000
0xa803346e5000
0xa80333cda000
0xa803326c3000
0xa8032f632000
0xa8032f135000
0xa80332915000
0xa80334a9e000
0xa80334d62000
0xa80337088000
pranavhemanth@Pranavs-MacBook-Pro-M3 DFIR-S6 %
```

Command: python3 -c "import sys; sys.argv[0]='volatility3'; from volatility3.cli import main; main()" -f image.mem windows.registry.printkey

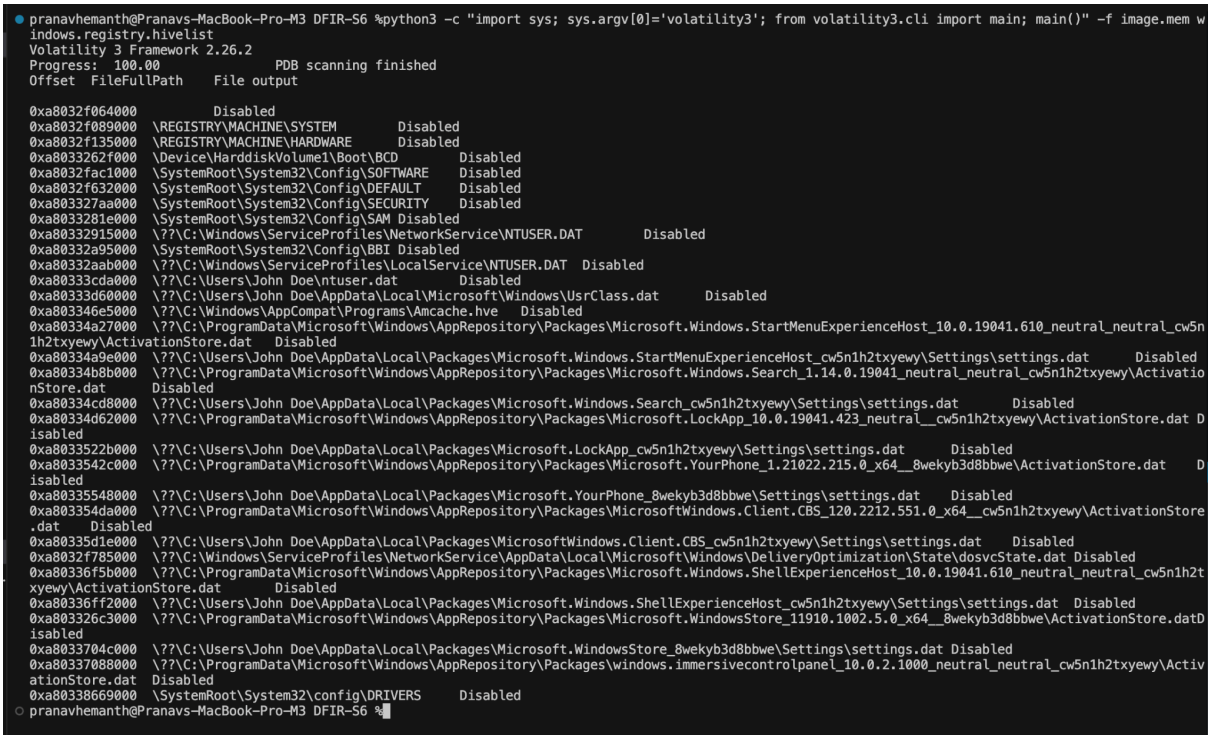
```
pranavhemanth@Pranavs-MacBook-Pro-M3 DFIR-S6 %python3 -c "import sys; sys.argv[0]='volatility3'; from volatility3.cli import main; main()" -f image.mem w
indows.registry.printkey
Volatility 3 Framework 2.26.2
Progress:  100.00               PDB scanning finished
Last Write Time Hive Offset     Type    Key     Name    Data    Volatile

2021-04-30 17:45:56.000000 UTC  0xa8032f064000  Key     [NONAME]        A       N/A     False
2021-04-30 17:51:19.000000 UTC  0xa8032f064000  Key     [NONAME]        MACHINE N/A     False
2021-04-30 17:39:47.000000 UTC  0xa8032f064000  Key     [NONAME]        USER    N/A     False
2021-04-30 17:51:18.000000 UTC  0xa8032f064000  Key     [NONAME]        WC      N/A     False
2019-12-07 09:15:08.000000 UTC  0xa8032f089000  Key     \REGISTRY\MACHINE\SYSTEM        ActivationBroker        N/A     False
2019-12-07 09:15:07.000000 UTC  0xa8032f089000  Key     \REGISTRY\MACHINE\SYSTEM        ControlSet001   N/A     False
2021-04-30 00:59:52.000000 UTC  0xa8032f089000  Key     \REGISTRY\MACHINE\SYSTEM        DriverDatabase  N/A     False
2021-04-30 12:39:38.000000 UTC  0xa8032f089000  Key     \REGISTRY\MACHINE\SYSTEM        HardwareConfig  N/A     False
2019-12-07 09:15:07.000000 UTC  0xa8032f089000  Key     \REGISTRY\MACHINE\SYSTEM        Input   N/A     False
2019-12-07 09:49:38.000000 UTC  0xa8032f089000  Key     \REGISTRY\MACHINE\SYSTEM        Keyboard Layout N/A     False
2019-12-07 09:15:07.000000 UTC  0xa8032f089000  Key     \REGISTRY\MACHINE\SYSTEM        Maps    N/A     False
2021-04-30 01:16:39.000000 UTC  0xa8032f089000  Key     \REGISTRY\MACHINE\SYSTEM        MountedDevices  N/A     False
2019-12-07 09:15:07.000000 UTC  0xa8032f089000  Key     \REGISTRY\MACHINE\SYSTEM        ResourceManager N/A     False
2019-12-07 09:15:07.000000 UTC  0xa8032f089000  Key     \REGISTRY\MACHINE\SYSTEM        ResourcePolicyStore     N/A     False
2021-04-30 12:39:38.000000 UTC  0xa8032f089000  Key     \REGISTRY\MACHINE\SYSTEM        RNG     N/A     False
2019-12-07 09:15:07.000000 UTC  0xa8032f089000  Key     \REGISTRY\MACHINE\SYSTEM        Select  N/A     False
2021-04-30 01:16:39.000000 UTC  0xa8032f089000  Key     \REGISTRY\MACHINE\SYSTEM        Setup   N/A     False
2019-12-07 09:15:07.000000 UTC  0xa8032f089000  Key     \REGISTRY\MACHINE\SYSTEM        Software        N/A     False
2019-12-07 09:15:07.000000 UTC  0xa8032f089000  Key     \REGISTRY\MACHINE\SYSTEM        State   N/A     False
2021-04-25 19:27:29.000000 UTC  0xa8032f089000  Key     \REGISTRY\MACHINE\SYSTEM        WaaS    N/A     False
2021-04-30 17:39:51.000000 UTC  0xa8032f089000  Key     \REGISTRY\MACHINE\SYSTEM        WPA     N/A     False
2021-04-30 12:39:38.000000 UTC  0xa8032f089000  Key     \REGISTRY\MACHINE\SYSTEM        CurrentControlSet       N/A     True
2021-04-30 12:39:38.000000 UTC  0xa8032f135000  Key     \REGISTRY\MACHINE\HARDWARE      ACPI    N/A     False
2021-04-30 12:39:38.000000 UTC  0xa8032f135000  Key     \REGISTRY\MACHINE\HARDWARE      DESCRIPTION     N/A     False
2021-04-30 12:39:44.000000 UTC  0xa8032f135000  Key     \REGISTRY\MACHINE\HARDWARE      DEVICEMAP       N/A     False
2021-04-30 12:39:38.000000 UTC  0xa8032f135000  Key     \REGISTRY\MACHINE\HARDWARE      RESOURCEMAP     N/A     True
2021-04-30 12:39:43.000000 UTC  0xa8033262f000  Key     \Device\HarddiskVolume1\Boot\BCD        Description     N/A     False
2021-04-30 12:39:43.000000 UTC  0xa8033262f000  Key     \Device\HarddiskVolume1\Boot\BCD        Objects N/A     False
2021-04-30 01:03:25.000000 UTC  0xa8032fac1000  Key     \SystemRoot\System32\Config\SOFTWARE    7-Zip   N/A     False
2021-04-29 16:15:09.000000 UTC  0xa8032fac1000  Key     \SystemRoot\System32\Config\SOFTWARE    AdoptOpenJDK    N/A     False
2021-04-30 01:20:14.000000 UTC  0xa8032fac1000  Key     \SystemRoot\System32\Config\SOFTWARE    Classes N/A     False
2019-12-07 09:52:45.000000 UTC  0xa8032fac1000  Key     \SystemRoot\System32\Config\SOFTWARE    Clients N/A     False
2021-04-25 17:29:52.000000 UTC  0xa8032fac1000  Key     \SystemRoot\System32\Config\SOFTWARE    CVSM    N/A     False
2019-12-07 09:15:15.000000 UTC  0xa8032fac1000  Key     \SystemRoot\System32\Config\SOFTWARE    DefaultUserEnvironment  N/A     False
2021-04-29 16:03:59.000000 UTC  0xa8032fac1000  Key     \SystemRoot\System32\Config\SOFTWARE    FileZilla 3     N/A     False
2019-12-07 09:15:15.000000 UTC  0xa8032fac1000  Key     \SystemRoot\System32\Config\SOFTWARE    Google  N/A     False
2019-12-07 09:15:15.000000 UTC  0xa8032fac1000  Key     \SystemRoot\System32\Config\SOFTWARE    Intel   N/A     False
2021-04-30 17:39:47.000000 UTC  0xa8032fac1000  Key     \SystemRoot\System32\Config\SOFTWARE    Microsoft       N/A     False
2019-12-07 09:15:10.000000 UTC  0xa8032fac1000  Key     \SystemRoot\System32\Config\SOFTWARE    ODBC    N/A     False
2019-12-07 09:15:10.000000 UTC  0xa8032fac1000  Key     \SystemRoot\System32\Config\SOFTWARE    OEM     N/A     False
2020-11-19 02:53:37.000000 UTC  0xa8032fac1000  Key     \SystemRoot\System32\Config\SOFTWARE    OpenSSH N/A     False
2021-04-25 18:07:02.000000 UTC  0xa8032fac1000  Key     \SystemRoot\System32\Config\SOFTWARE    Oracle  N/A     False
2019-12-07 09:15:10.000000 UTC  0xa8032fac1000  Key     \SystemRoot\System32\Config\SOFTWARE    Partner N/A     False
2019-12-07 09:17:32.000000 UTC  0xa8032fac1000  Key     \SystemRoot\System32\Config\SOFTWARE    Policies        N/A     False
2021-04-25 18:44:58.000000 UTC  0xa8032fac1000  Key     \SystemRoot\System32\Config\SOFTWARE    RegisteredApplications  N/A     False
2019-12-07 09:15:10.000000 UTC  0xa8032fac1000  Key     \SystemRoot\System32\Config\SOFTWARE    -       N/A     False
2021-04-30 12:39:44.000000 UTC  0xa8032fac1000  Key     \SystemRoot\System32\Config\SOFTWARE    WOW6432Node     N/A     False
2020-11-19 07:30:28.000000 UTC  0xa8032f632000  Key     \SystemRoot\System32\Config\DEFAULT     Console N/A     False
2019-12-07 09:49:38.000000 UTC  0xa8032f632000  Key     \SystemRoot\System32\Config\DEFAULT     Control Panel   N/A     False
2019-12-07 09:15:07.000000 UTC  0xa8032f632000  Key     \SystemRoot\System32\Config\DEFAULT     Environment     N/A     False
2019-12-07 09:15:07.000000 UTC  0xa8032f632000  Key     \SystemRoot\System32\Config\DEFAULT     EUDC    N/A     False
2021-04-25 19:29:30.000000 UTC  0xa8032f632000  Key     \SystemRoot\System32\Config\DEFAULT     Keyboard Layout N/A     False
2020-11-19 07:34:31.000000 UTC  0xa8032f632000  Key     \SystemRoot\System32\Config\DEFAULT     Printers        N/A     False
2020-11-19 07:30:54.000000 UTC  0xa8032f632000  Key     \SystemRoot\System32\Config\DEFAULT     Software        N/A     False
2020-11-19 07:32:11.000000 UTC  0xa8032f632000  Key     \SystemRoot\System32\Config\DEFAULT     System  N/A     False
```

37.89MB

Windows registry hives stored in memory were identified using Volatility 3's windows.registry.hivelist and windows.registry.hivescan plugins. These plugins locate

loaded and residual registry hive structures within memory, enabling investigators to identify and extract registry hives such as SYSTEM, SAM, SOFTWARE, and user hives for further offline analysis.

10. Identify Persistence Mechanisms via Registry
    Question: How can you identify malware persistence mechanisms using registry Run keys stored in memory?
    Answer:
    Command: python3 -c "import sys; sys.argv[0]='volatility3'; from volatility3.cli import main; main()" -f image.mem windows.registry.hivelist



```
pranavhemanth@Pranavs-MacBook-Pro-M3 DFIR-S6 %python3 -c "import sys; sys.argv[0]='volatility3'; from volatility3.cli import main; main()" -f image.mem w
indows.registry.hivelist
Volatility 3 Framework 2.26.2
Progress:  100.00               PDB scanning finished
Offset  FileFullPath    File output

0xa8032f064000          Disabled
0xa8032f089000  \REGISTRY\MACHINE\SYSTEM        Disabled
0xa8032f135000  \REGISTRY\MACHINE\HARDWARE      Disabled
0xa8033262f000  \Device\HarddiskVolume1\Boot\BCD        Disabled
0xa8032fac1000  \SystemRoot\System32\Config\SOFTWARE    Disabled
0xa8032f632000  \SystemRoot\System32\Config\DEFAULT     Disabled
0xa803327aa000  \SystemRoot\System32\Config\SECURITY    Disabled
0xa8033281e000  \SystemRoot\System32\Config\SAM Disabled
0xa80332915000  \??\C:\Windows\ServiceProfiles\NetworkService\NTUSER.DAT        Disabled
0xa80332a95000  \SystemRoot\System32\Config\BBI Disabled
0xa80332aab000  \??\C:\Windows\ServiceProfiles\LocalService\NTUSER.DAT  Disabled
0xa80333cda000  \??\C:\Users\John Doe\ntuser.dat         Disabled
0xa80333d60000  \??\C:\Users\John Doe\AppData\Local\Microsoft\Windows\UsrClass.dat      Disabled
0xa803346e5000  \??\C:\Windows\AppCompat\Programs\Amcache.hve   Disabled
0xa80334a27000  \??\C:\ProgramData\Microsoft\Windows\AppRepository\Packages\Microsoft.Windows.StartMenuExperienceHost_10.0.19041.610_neutral_neutral_cw5n
1h2txyewy\ActivationStore.dat   Disabled
0xa80334a9e000  \??\C:\Users\John Doe\AppData\Local\Packages\Microsoft.Windows.StartMenuExperienceHost_cw5n1h2txyewy\Settings\settings.dat      Disabled
0xa80334b8b000  \??\C:\ProgramData\Microsoft\Windows\AppRepository\Packages\Microsoft.Windows.Search_1.14.0.19041_neutral_neutral_cw5n1h2txyewy\Activatio
nStore.dat      Disabled
0xa80334cd8000  \??\C:\Users\John Doe\AppData\Local\Packages\Microsoft.Windows.Search_cw5n1h2txyewy\Settings\settings.dat       Disabled
0xa80334d62000  \??\C:\ProgramData\Microsoft\Windows\AppRepository\Packages\Microsoft.LockApp_10.0.19041.423_neutral__cw5n1h2txyewy\ActivationStore.dat D
isabled
0xa8033522b000  \??\C:\Users\John Doe\AppData\Local\Packages\Microsoft.LockApp_cw5n1h2txyewy\Settings\settings.dat      Disabled
0xa8033542c000  \??\C:\ProgramData\Microsoft\Windows\AppRepository\Packages\Microsoft.YourPhone_1.21022.215.0_x64__8wekyb3d8bbwe\ActivationStore.dat    D
isabled
0xa80335548000  \??\C:\Users\John Doe\AppData\Local\Packages\Microsoft.YourPhone_8wekyb3d8bbwe\Settings\settings.dat    Disabled
0xa803354da000  \??\C:\ProgramData\Microsoft\Windows\AppRepository\Packages\MicrosoftWindows.Client.CBS_120.2212.551.0_x64__cw5n1h2txyewy\ActivationStore
.dat    Disabled
0xa80335d1e000  \??\C:\Users\John Doe\AppData\Local\Packages\MicrosoftWindows.Client.CBS_cw5n1h2txyewy\Settings\settings.dat    Disabled
0xa8032f785000  \??\C:\Windows\ServiceProfiles\NetworkService\AppData\Local\Microsoft\Windows\DeliveryOptimization\State\dosvcState.dat Disabled
0xa80336f5b000  \??\C:\ProgramData\Microsoft\Windows\AppRepository\Packages\Microsoft.Windows.ShellExperienceHost_10.0.19041.610_neutral_neutral_cw5n1h2t
xyewy\ActivationStore.dat       Disabled
0xa80336ff2000  \??\C:\Users\John Doe\AppData\Local\Packages\Microsoft.Windows.ShellExperienceHost_cw5n1h2txyewy\Settings\settings.dat  Disabled
0xa803326c3000  \??\C:\ProgramData\Microsoft\Windows\AppRepository\Packages\Microsoft.WindowsStore_11910.1002.5.0_x64__8wekyb3d8bbwe\ActivationStore.datD
isabled
0xa8033704c000  \??\C:\Users\John Doe\AppData\Local\Packages\Microsoft.WindowsStore_8wekyb3d8bbwe\Settings\settings.dat Disabled
0xa80337088000  \??\C:\ProgramData\Microsoft\Windows\AppRepository\Packages\windows.immersivecontrolpanel_10.0.2.1000_neutral_neutral_cw5n1h2txyewy\Activ
ationStore.dat  Disabled
0xa80338669000  \SystemRoot\System32\config\DRIVERS     Disabled
pranavhemanth@Pranavs-MacBook-Pro-M3 DFIR-S6 %
```

Command: python3 -c "import sys; sys.argv[0]='volatility3'; from volatility3.cli import main; main()" -f image.mem windows.registry.printkey --key "Software\Microsoft\Windows\CurrentVersion\Run"

Malware persistence mechanisms were identified by analyzing registry Run and RunOnce keys stored in memory using Volatility 3's registry analysis plugins. The windows.registry.hivelist plugin was used to locate loaded registry hives, and the windows.registry.printkey plugin was used to examine Run key locations commonly abused for persistence. Suspicious startup entries indicate potential malware persistence.

Conclusion

This lab demonstrates the use of Volatility 3 to analyze a Windows memory dump and extract critical forensic artifacts such as system information, running processes, network connections, user activity, and persistence mechanisms. These techniques are essential for incident response and memory forensics investigations.